

Near Neighbor Helping Neighbor: Local Giving and the Geography of Charity*

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Abstract

Does the decentralized, community funding structure of private charities create a gap between where high-capacity food pantries are located and where resource demand is highest? In this paper, I find that food pantry customers living in high-income, rural neighborhoods live within 20 miles of 40 more food pantries than customers in low-income neighborhoods. Customers in low-income neighborhoods also face more restrictions, such as limits on the number of visits allowed per month. Using observed customer behavior, I estimate the value of each food pantry visit based on characteristics such as flexible distribution times and availability of food. I then show that customers in high-income neighborhoods receive higher per-visit benefits than those in low-income neighborhoods. In a counterfactual simulation, I show that redistributing funding to improve pantries in low-income areas increases consumer surplus by 3.3% overall, with welfare gains of 4–5% for customers in the lowest-income neighborhoods, equivalent to nearly one additional meal per visit.

JEL Codes: D64; H23; I32; L31; Q18

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1 Introduction

Since the early 1980s, reductions in federal support for social programs have shifted responsibility for social service provision toward charitable organizations. In a speech to religious leaders, Reagan said he believed it was a “good thing for the soul of the country to encourage people to get involved and accept more direct responsibility for one another’s health, happiness, and well-being, rather than leaving it to the bureaucracy” (Reagan 1982). However, the provision of charitable goods relies on a decentralized funding structure dependent on the preferences of private donors, foundations, and volunteers. Because private support is likely tied to localized preferences, charitable capacity can fail to align with the geographic distribution of need.

This paper explores the geographic distribution and welfare implications of charitable capacity in rural communities, where the provision of services relies heavily on local resources. Researchers show that donors prefer to support organizations in their local community (Card, Hallock, and Moretti 2010; DellaVigna, List, and Malmendier 2012; List 2011; Blackwell and McKee 2003). Because private, charitable organizations rely on localized support from donors and volunteers, the level of service that these organizations can provide and the number of customers they can serve are tied to the economic capacity of their surrounding geographic area. If higher-income communities are better equipped to fund charitable services, how does this affect the geographic distribution of high-capacity, charitable services, and would social welfare improve if donations were redistributed?

While this paper focuses on food pantries, these questions apply broadly to other charitable organizations—including low-cost childcare services, affordable housing, community meals, public radio, and healthcare—where the capacity to provide services depends primarily on donors and volunteers. Food pantries are particularly equipped to answer this question because they are small, locally operated, and some pantries rely entirely on community support. Unlike national charities that support research or advocacy, food pantries provide resources to their local community, and some pantries restrict their customers’ eligibility by geographic location, like zip code or school district. This location dependence creates a spatial mismatch between resource availability and need.

Food pantries are ubiquitous and heavily utilized. More than 60,000 food pantries are affiliated with Feeding America,¹ and over 50 million people—nearly 15% of the U.S. population—used a food pantry in 2023 (Feeding America 2024). By comparison, the Supplemental Nutrition Assistance Program (SNAP) served 12.6% of the population in 2023 (U.S. Department of Agriculture, Economic Research Service 2024). This difference partly reflects eligibility: food pantries often serve households with incomes up to 300% of the federal poverty line, while SNAP eligibility is typically limited to households below 130%. However, SNAP provides regular, monthly benefits that households can use

¹Feeding America is a national organization that coordinates and supports independent charitable food programs, including food banks and food pantries. Affiliation with Feeding America is optional, and organizations apply to become a Feeding America partner pantry, typically coordinated through a Feeding America affiliated food bank.

to purchase groceries, whereas food pantry access is episodic and varies widely across locations and over time.

To understand the relationship between charitable capacity and the demand for charitable services, I use a novel data set from the Mid-Ohio Food Collective (MOFC), a large food bank in central Ohio that provides food to pantries in 20 counties. These data include household-level details on food pantry utilization for over 100,000 food pantry customers, including the date and time the customer visited a pantry and the distance between the customer's home and the food pantry.

Focusing on 19 rural counties in the network, I document spatial differences in food pantry locations. While fewer food pantry customers live in high-income neighborhoods,² these customers have over 40 more food pantries within 20 miles of their residence when compared to customers living in low-income neighborhoods. The characteristics of the food pantries near high- versus low-income neighborhoods also vary; a larger share of customers in low-income neighborhoods do not live near any pantry that allows flexible access. For example, though all pantries in the MOFC network must provide at least three days' worth of food each month, many pantries restrict the quantity of food provided per household through limits on the number of times a customer can visit within a month. Other pantries allow unlimited visits. 72% of customers in low-income neighborhoods live near at least one pantry without restrictions on visit frequency compared to 96% of customers in high-income neighborhoods.

To determine if differential access leads to disparities in the value that customers derive from visiting a food pantry, I use a random utility model allowing for heterogeneous preferences. In the model, customers either choose to source their meals from any of the food pantries within 20 miles of their home or from a grocery retailer. While visiting the pantry is free, customers face travel costs. They choose the pantry that provides the greatest benefit, such as more food or flexible distribution times, given the distance they must travel.

I find systematic differences in food pantry characteristics across neighborhoods. Pantries in high-income areas offer characteristics that customers value leading to a better experience for customers in these neighborhoods. I also find that customers in the highest-income neighborhoods are more sensitive to travel costs, yet these customers are willing to travel farther than customers in low-income neighborhoods to access their preferred pantry. Together, these results suggest that customers in high-income neighborhoods receive greater per-visit benefits than those in low-income neighborhoods.

Given that high-capacity pantries tend to be concentrated in high-income areas, I investigate whether redistributing funding would improve welfare for low-income customers. Using the preferences derived from the random utility model, I estimate the counterfactual change in consumer welfare if a portion of funds supporting pantries in high-income neighborhoods were redistributed to

²All food pantry customers in the data used for this project are self-reported to be below 200% of the federal poverty line. Therefore, a food pantry customer living in a high-income neighborhood is not a high-income individual.

pantries in low-income neighborhoods. Under the counterfactual funding distribution, I find that customer welfare increases in low-income areas by 4–5% and decreases in high-income areas by 1–2.5%, resulting in overall customer welfare increases of 3.3% over the decentralized funding structure. My findings suggest that consumer welfare could improve by redistributing food pantry funding, which implies that decentralized, preference-based allocations result in a distribution of resources that is misaligned with customer needs. While redistributing funding across independent organizations is not feasible, the counterfactual analysis highlights welfare gains from reallocating external funding, such as funding from private foundations, toward under-resourced organizations in high-demand areas.

My work lies at the intersection of literature on spatial inequities, charitable giving, and social service provision. Research on spatial inequities demonstrates how neighborhood characteristics shape access to resources. Early literature focused on the effects of segregation on amenities, housing, and access to labor markets (Cutler, Glaeser, and Vigdor 1999; Brueckner, Thisse, and Zenou 1999; Gobillon, Selod, and Zenou 2007; Waldfogel 2008). More recent work finds that low-income and predominantly Black neighborhoods face higher costs and reduced access to infrastructure and markets. For example, these neighborhoods experience rougher roads (Currier, Glaeser, and Kreindler 2023) and have more limited access to rooftop solar photovoltaic markets (Dorsey and Wolfson 2024). This literature highlights how residents of low-income neighborhoods, who are less likely to have expendable income, often are faced with higher costs in terms of local infrastructure and accessing other amenities. My work adds to this literature by demonstrating that individuals living in low-income, rural areas are also less likely to have access to well-resourced food pantries

This work connects to the literature on charitable giving by documenting the downstream spatial consequences of localized donor preferences on organizational capacity.³ In addition to research showing that donors and volunteers prefer to support local organizations, List (2011) finds that higher-income households have a higher probability of giving to charity than lower-income households, and give more than lower-income households conditional on giving to charity. This finding suggests that charities in areas with fewer high-income households may find it more difficult to attract donors and may receive less money per donation. Structural models further suggest that nonprofits are only economically viable in markets with a minimum number of potential donors, and that this threshold varies by nonprofit sector, such as performing arts, community health, or museums (Gayle, Harrison, and Thornton 2017). These findings motivate my analysis of the resulting spatial inequities in charitable supply, allowing me to evaluate how disparities in capacity ultimately determine client access and welfare across neighborhoods.

Other nonprofit characteristics influence a charitable organization’s ability to attract donors. A

³In addition to research on donor preferences, there is extensive literature on the fundraising efforts of charitable organizations. For example, see Name-Correa and Yildirim (2013); Andreoni and Payne (2003); and Andreoni and Payne (2011).

highly visible increase in need, like a natural disaster, can lead to increases in giving (Deryugina and Marx 2021) while subtler changes, such as rising food insecurity, are less responsive (Karol 2023). Donor behavior is also influenced by group identity. Using a public goods experiment that varies the benefit of contributing to a good that is exclusive to one's own group versus a non-excludable, global public good, Blackwell and McKee (2003) find that participants seem to have a bias to support the group good, meant to represent local, public goods. The authors find that this bias can be overcome if the global public good's benefit to society is sufficiently high. Along the same lines, Andreoni et al. (2016) find that charitable contributions decrease as racial, ethnic, and religious diversity increases. Together, these patterns suggest that donor preferences are partially driven by local ties and demographic characteristics. My work shows that prioritizing local organizations can create gaps between charitable resources and community needs.

Finally, this paper contributes to the literature on the role of food pantries in the larger food assistance system by quantifying the surplus derived from a food pantry visit. While food banks and food pantries are a part of the nationwide response to food insufficiency, most of the research on food pantries relies on surveys or qualitative data. In addition, the existing work emphasizes household characteristics associated with food pantry utilization or the nutritional value of food provided (Byrne and Just 2022). In contrast, this paper uses detailed, visit-level data to estimate customer preferences and quantify per-visit benefits.

Research using administrative data from food banks and food pantries is limited. Byrne and Just (2021) use administrative data from a large, urban food bank to document cyclical food pantry utilization that corresponds to when SNAP benefits run out toward the end of the month. In another study, Byrne and Just (2023) use a travel costs approach to estimate that each food pantry visit is worth between \$40 and \$60. To my knowledge, my work is the first, large-scale study of cross-county food pantry utilization and demand, and the first focusing on rural food pantry access.

The remainder of this paper proceeds as follows: Section 2 explains the charitable food sector in the United States and in the focal region in rural Ohio. Section 3 describes the data used in the analysis and Section 4 explains the conceptual framework behind the spatial mismatch of food pantry service and the need for charitable food. Section 5 provides a descriptive analysis of food pantry access among households in high- and low-income neighborhoods. Section 6 details the demand estimation model used to quantify customer preferences. Section 7 describes the results and counterfactual analysis. Section 8 concludes.

2 Background and Environment

2.1 Food Banks and Food Pantries

Private, charitable food assistance likely pre-dates most other public assistance programs. For instance, the Salvation Army adopted the emblematic red kettles to collect their Christmas donations as a nod to Count Rumford, who is credited with establishing the first soup kitchen in the late 1700s (Bramen 2010). The term ‘breadline’ was coined in the late 1800s by Louis Fleischmann, a baker who offered to feed hungry people who came to his bakery in New York City. The modern U.S. food banking system was established much later in 1967 by John van Hengel, who founded the St. Mary’s Food Bank in Arizona (Byrne and Just 2022; Mook, Murdock, and Gundersen 2020). In this model of food banking, food is donated to or purchased by food banks and then distributed to food pantries.

In many ways, food banks and their network of food pantries operate like wholesale distributors and grocery retailers, where independently owned retailers purchase food from a distributor. The food bank stores the food in a large-scale warehouse and food pantries, which are independent organizations, purchase food from the food bank to distribute to the public. The pantries receive some food, such as fresh produce, either free of charge or for a small fee to cover storage and delivery. Pantries purchase other products at-cost, allowing the food bank to pass on savings from high-volume purchases.

Charitable food programs grew in the 1980s due to a combination of factors (Poppendieck 1999). Economic stagnation in the 1970s led to high poverty rates that continued to climb into the 1980s. At the same time, the Reagan Administration implemented cuts to social programs, which led many families to turn to charitable assistance to make ends meet. Finally, reports of a government-owned stock of dairy surplus in the face of widespread hunger led to the federal institutionalization of commodities distribution through the Temporary Emergency Food Assistance Program in 1981.

The initial intention of the program was to be a temporary method of distributing government agricultural surpluses, but in 1990, Congress removed ‘Temporary’ from the name of the program, renaming it The Emergency Food Assistance Program (TEFAP). Since then, Congress appropriates funds to purchase just under \$500 million in USDA foods annually to be distributed through TEFAP, along with limited funding for storage and distribution.⁴ TEFAP also acquires surplus food from agricultural markets to distribute throughout the year, including \$955 million in agricultural surpluses in fiscal year 2023 (U.S. Department of Agriculture, Food and Nutrition Service 2024). Foods available through TEFAP include fresh and canned fruits and vegetables (apples, peaches, potatoes, peas), milk, eggs, pasta, cereal, dry and canned beans, rice, peanut butter, nuts, and cheese (U.S. Department of Agriculture, Food and Nutrition Service 2025b).

State governing agencies, such as Job and Family Services in Ohio, distribute TEFAP products to

⁴Funding for administrative costs from TEFAP totaled \$80 million in fiscal year 2024 (U.S. Department of Agriculture, Food and Nutrition Service 2024).

food banks authorized to distribute USDA foods. State-level funds also support food banks through their own programs. The Ohio Community+Agriculture+Nutrition, or Ohio CAN,⁵ provides funding for food banks to purchase food from local farmers. Ohio also has several programs similar to TEFAP including the Ohio Food Program, which uses state funds to provide shelf-stable products to food banks, and the Ohio Agricultural Clearance Program, which makes agricultural surplus from Ohio farmers available to food banks. The food banks distribute to their food pantry network either free of charge or at a steep discount.

In addition to government funding, private financial contributions from individuals, corporations, and private foundations subsidize the operating costs of food banks, including staff, food storage and distribution, and food purchases. Donations from private sources can make up a substantial portion of the food bank's budget. For instance, the Mid-Ohio Food Collective (MOFC) brought in \$19.3 million in donations from individuals and corporations in 2023, nearly 60% of the non-food budget.⁶

Unlike food banks, which receive some federal and state support, food pantries rely almost entirely on private funding or funding from local governments. Well-funded pantries may own facilities equipped with storage and refrigeration, allowing for regular distribution of a variety of products. Other organizations may borrow space, like a church fellowship hall or school gymnasium, and must distribute all products during limited hours due to storage limitations. This often forces pantries to purchase less food to avoid leftovers.⁷

Typically, food pantries operate as a part of a local nonprofit organization, a church, or a branch of a larger nonprofit organization. Many food pantries are all-volunteer organizations, and therefore can only offer food if enough volunteers are available to assist with distribution. These organizations also need devoted volunteers willing to manage the logistical needs of the pantry, such as ordering food, coordinating pick-up and delivery, marketing the services, and fundraising. Larger organizations with full-time staff still rely on volunteer labor to operate. For instance, full-time staff may coordinate a group of volunteers who then pack and distribute the food to the community. In 2022, food pantries in the MOFC network that were large enough to file a Form 990 reported a median of 10 staff members and 192 volunteers.

⁵Though Ohio CAN was active during the time period studied in this paper, it has since been discontinued due to funding cuts.

⁶For context, Food Bank News (2023), a publication that ranks US food banks by revenue, ranked MOFC 40th of the 300 US food banks studied, the second-largest by revenue in Ohio behind the Greater Cleveland Food Bank.

⁷One way of addressing storage issues is offering pop-up distribution sites, known as "mobile food pantries" or "produce markets," in areas known to face food access challenges. Such pop-up distribution sites may be situated in low-income housing communities, community centers, library parking lots, or other centralized locations. Many food pantries coordinate mobile distribution, and the food bank studied in this paper also intentionally organizes several produce markets throughout the region to target underserved areas. A food bank employee trucks in food and stays until the end of the distribution to return undistributed food to the food bank, which circumvents the issue of storage.

2.2 The Mid-Ohio Food Collective

MOFC is a large nonprofit organization that operates the Mid-Ohio Food Bank and is the exclusive food bank for over 700 agencies in 20 of the 88 counties in Ohio. Most of these agencies are food pantries, but some are senior centers, community meal sites, daycares, and other organizations.

This paper specifically looks at the 19 rural counties served by MOFC and the over 200 pantries that serve rural customers, as shown in Figure 1. I exclude customers living in Franklin County, which is mostly urban and home to Columbus. I focus on rural counties because, in general, rural counties face higher poverty and food insecurity rates than their urban counterparts (Haynes-Maslow, Andress, Byker-Shanks, et al. 2020), and yet urban areas receive most of the academic attention. Feeding America (2025) finds that 86% of counties with high food insecurity are rural. MOFC calculates an internal ‘Missing Meal Index’ using data from the American Community Survey and the Bureau of Labor Statistics to quantify the number of meals that members of the MOFC service area skip due to economic hardship.⁸ They estimate that individuals below 200% of the federal poverty line skip 3.56 meals per week on average in the 20 county footprint. In Franklin County, the Missing Meal Index is among the lowest at 3.21 meals per week, while the index is above four meals per week in almost all rural counties. In addition, urban and rural areas in Ohio face different challenges in supporting low-income households, such as variation in access to public transportation, employment opportunities, and public services, making it difficult to compare access in urban and rural areas. I therefore leave research on urban food pantry access to future work.

In spite of the variation between pantries, all pantries within the MOFC network receive training and support and meet requirements set forth by MOFC, the USDA, and Job and Family Services (JFS). JFS requires households to earn less than 200% of the federal poverty line to be eligible for food pantry services, in contrast with SNAP benefits, which are available to individuals up to 130% of the federal poverty line. Additionally, while SNAP benefits involve a long application process, work and asset requirements, and potentially long wait times (Herd and Moynihan 2019), customers can come to the food pantry and take food home in the same day. Many states allow individual organizations to determine if they would like to request proof of income, but Ohio is among the few states that are “self-declaration states,” which means that food pantries and other agencies are forbidden from asking their customers to show proof of income in exchange for food (Feeding America Action 2020). This policy reduces barriers to entry into the food pantry system, allowing customers to participate with minimal administrative burden.

⁸To arrive at this figure, MOFC calculates the total number of meals that families under 200% of the federal poverty line need, assuming three meals per person, and the cost of those meals using Feeding America’s Average Meal cost, which is adjusted for county-level food prices and taxes. They then subtract out meals purchased by the individuals using the Bureau of Labor Statistics Consumer Expenditure Survey, meals provided by the government (SNAP, WIC, NSLP, etc.) using average expenditure and utilization rates, and food provided by charitable resources using their own data. The number of missing meals is the number of meals not covered by the individual, government, or charitable programs.

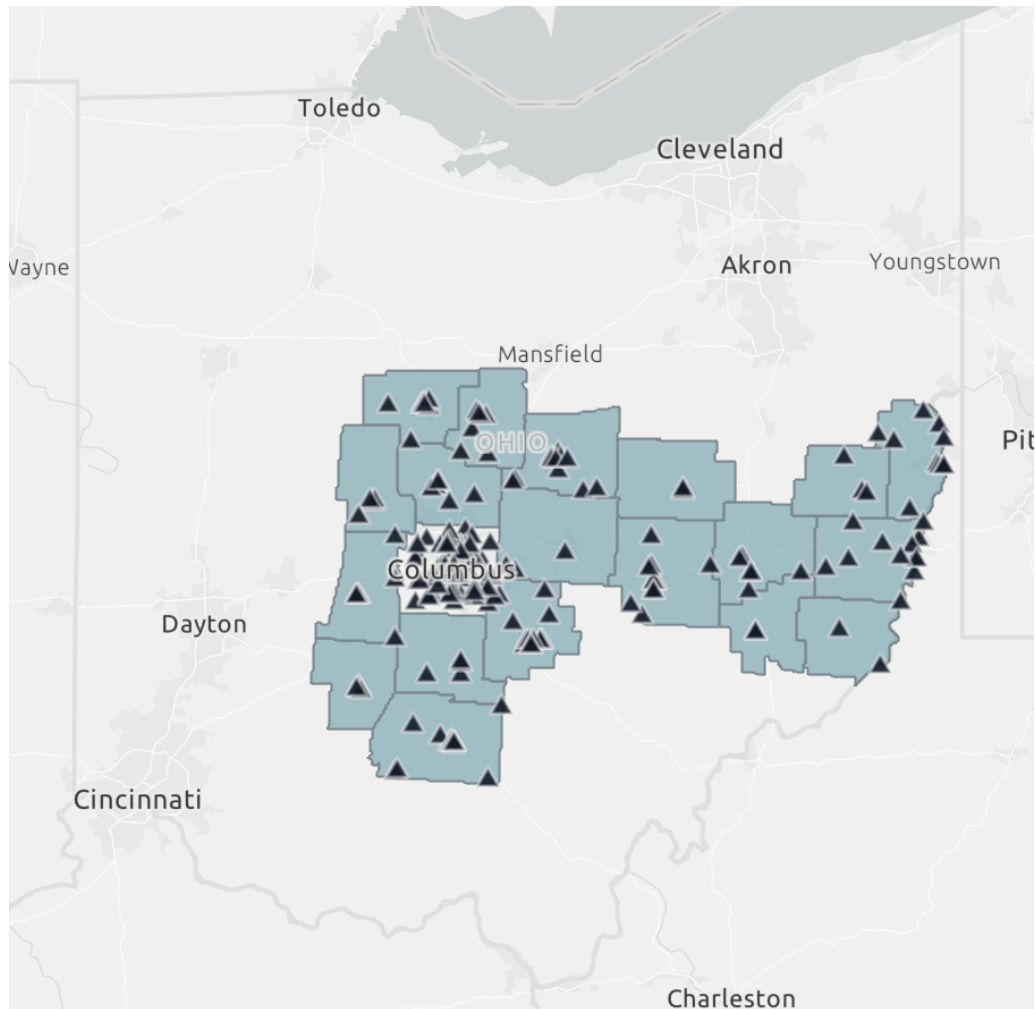


Figure 1: Map of MOFC Rural Counties & Pantry Locations

Notes: The map displays the rural counties served by MOFC and marks the locations of the over 200 food pantries accessible to rural households.

In addition to enforcing government guidelines, MOFC ensures that the pantries within the network adhere to additional requirements, including food safety training, food storage guidelines, civil rights training, and reporting requirements (Mid-Ohio Foodbank 2018). MOFC imposes a minimum service requirement of three days' worth of groceries at least once per calendar month, and strongly encourages pantries to provide five days' worth of groceries if possible. Although MOFC allows food pantries to set geographic service area restrictions, they cannot require that their customers participate in activities unrelated to food service provision to receive food. In other words, pantries cannot require that their customers participate in religious services, sign petitions, or donate to a cause in exchange for food from MOFC. To enforce these policies, MOFC randomly audits the food pantries within the network and evaluates customer complaints.

Although MOFC sets network-wide standards for eligibility and service provision, households' realized access depends on the characteristics of the individual pantries they visit. Food pantries distribute food in-kind, and the quantity and variety available are shaped by pantry-level constraints such as inventory, hours of operation, geographic service areas, and limits on the amount of food provided per household. These constraints typically arise from limited resources and high demand rather than formal eligibility rules. As a result, households may not fully meet their needs at a single pantry. MOFC does not restrict households to one location, however, and customers may therefore visit multiple pantries within the network—including more than one in the same day—to obtain additional food.

3 Data

3.1 Pantry Utilization and Order Data

The primary data come from the MOFC client-management system. When a customer first visits a pantry, they fill out an intake form, required by JFS for charitable food program eligibility. The form includes a section where households agree that their household income falls below 200% of the federal poverty line, and report their address, household size, and the number of children (birth to 17), adults, and adults over 60. Upon first visiting a food pantry in the MOFC network, the pantry captures this information through the digital client-management system and assigns each household a unique household identifier. On subsequent visits, customers check in to the pantry and verify their information is up-to-date, or update their profile if necessary. The system records each visit and links all additional visits within the network.

The data used in this paper include the recorded household demographics extracted from the intake forms, the dates the customer visited a pantry, and a geocoded household location.⁹ The location

⁹Exact household locations are masked to prevent individual identification.

information allows me to match households with census tract data from the American Community Survey, which I use to define a neighborhood. Specifically, I take the median household income of each of the 65 census tracts in the 19 rural counties served by MOFC, and I divide the tracts into five income quintiles. While I conduct my primary analysis at the individual-level, I assign each individual in the sample a neighborhood income quintile that corresponds to the income quintile of the tract where they live. Because all food pantry customers are below 200% of the federal poverty line, the income quintile captures the economic resources and philanthropic capacity of the surrounding community rather than the customer's own household income. Identifying the neighborhood income quintile allows me to characterize how food pantry capacity varies with the economic and donor resources of the surrounding area.

I also obtain detailed data about each food pantry from MOFC, including the pantry's location, Employer Identification Number (EIN), and invoices for food orders from the food bank. The pantry's geocoded location allows me to calculate the Euclidean distance between a food pantry customer's home location and all food pantries. To indicate organization type and structure, MOFC provided the EINs for most organizations. I fill in missing EINs by hand by searching for the organization using the ProPublica EIN lookup, a search engine that scrapes data from the available tax filings from the Internal Revenue Service (IRS)(Suozzo et al. 2026).

I use the EINs to match agencies with their publicly-available Form 990 tax document. IRS requires a different version of the form based on the organizations gross receipts and total assets. Organizations with gross receipts over \$200,000 or total assets over \$500,000 file a Form 990, while organizations under these thresholds file a shorter, Form 990-EZ. The smallest organizations, with gross receipts under \$50,000, file a Form 990-N, known as a 'postcard,' which essentially confirms that the nonprofit is still actively conducting business under their EIN.

The National Center for Charitable Statistics (NCCS) compiles organizational data from IRS Forms 990 and 990-EZ, as well as a list of Form 990-N filers. However, NCCS data are limited to organizations that file electronically; because e-filing was not required prior to 2022, filings are missing for many organizations in earlier years. The dataset also excludes small organizations that are not required to file Form 990s. In addition, faith-based organizations, such as churches, are generally exempt from filing requirements, despite representing a substantial share of food pantries and other social service providers, particularly in rural areas. As a result, the data likely underrepresent smaller and faith-based providers, which may be disproportionately important in rural service provision.

Matching NCCS records to MOFC administrative data reveals a stark discrepancy: NCCS accounts for only 35% of the organizations actively providing food in the region. This gap suggests that studies relying solely on NCCS data (e.g., Ruffini, Öztürk, and Pekgün (2025) or Heflin and Harrington Meyer (2025)) may significantly under count small, community-based providers, skewing research toward larger organizations. By scraping individual IRS filings and matching them to local administrative

records, I construct a more comprehensive inventory of the charitable food landscape in rural Ohio, capturing the small providers that are often overlooked in national datasets.

Using NCCS and IRS data, I classify organizations by type, distinguishing among church-based pantries, independent nonprofit organizations, and branches of larger nonprofit organizations. Church-based organizations are exempt from filing IRS tax returns. For organizations without a unique Employer Identification Number (EIN), I use information from organizational websites or, when unavailable, from organization names (e.g., “Methodist Church Pantry”) to determine organizational status.

Branches are affiliate locations of larger nonprofit organizations that share a common EIN and file consolidated tax returns with the parent organization. For example, the Salvation Army operates several food pantries throughout the MOFC region. In the data, I observe separate food orders and customer visits for each location as if they are separate locations. However, these organizations do not appear as separate entities in IRS filing data. Forms 990 and 990-EZ additionally provide information on organizational operations, including staffing levels, volunteer participation, revenues from grants and donations, and expenditures on employee compensation, advertising, and rent.

Finally, I observe the types of food the pantries order from MOFC through their online ordering system. The online system includes the full inventory of food available for purchase, including the name of the food product (i.e., ‘Red Pepper Hummus’) and several categories to describe the food (i.e. fresh, junk food, canned, etc.). I access the invoices for all pantry orders within the MOFC ordering systems, including the quantity (in pounds) and description of food ordered, such as the name of the product and related food categories. I also obtain the date and time of the order. I aggregate the total number of pounds of food each pantry ordered per year, which I then divide by the number of visits to each pantry to proxy for the amount of food visitors could expect to receive at each visit.¹⁰

3.2 Summary Statistics

I focus on food pantry utilization between 2018 and 2023, when all pantries in the MOFC network used the client-management system. I also only include households with complete home location data.¹¹ Table 1 reports summary statistics about the sample of food pantry customers used for analysis. Overall, there are 102,403 unique households in rural counties that used a food pantry between 2018

¹⁰Although pantries can acquire food through donations from local retailers, MOFC manages the relationship between pantries and most large chain retailers. This means that retailers can donate food to the MOFC and then deliver it directly to the local food pantry that needs it instead of transporting it to MOFC for redistribution. In my conversations with MOFC, they stated that they surveyed the pantries in their network, and the pantries estimated that over 75% of their food stock came from MOFC. MOFC suspects that many pantries do not realize that food from chain retailers is also distributed through the food bank, and therefore the estimate of 75% likely underestimates the total amount of food supplied by MOFC.

¹¹Because home location data define the choice set and demand for food pantries in my empirical work, I exclude individuals experiencing homelessness.

and 2023. A majority of customers who visit the pantry are female, and the average household size is just under three individuals. Customers travel a one-way, Euclidean distance of 4.9 miles on average per visit.

Table 1: Customer Summary Statistics

	Mean	Standard deviation
<i>Panel A. Overview</i>		
Age at first visit	45.1	16.6
Total pantry options	26.5	30.7
Distance traveled per visit (one way, miles)	4.8	4.9
Household size	2.8	1.9
Children in household	0.9	1.3
Seniors in household	0.4	0.7
Adults (18-64) in household	1.5	1.2
<i>Panel B. Gender</i>		
% Male	30.5	
% Female	59.7	
Observations	102,403	

Notes: This table reports summary statistics about the population that used a food pantry in rural, MOFC counties between 2018 and 2023. Note that Male and Female do not add up to 100 percent as some pantry customers either did not report a gender or reported ‘Other.’ Total pantry options includes the count of pantries within a 20-mile radius of the customer’s home location. Average distance is the average, linear distance from customer’s home location to the pantry they visited. (Source: author’s calculations using MOFC food pantry utilization data.)

To estimate pantry access, I define each customer’s choice set, or the pantries the customer could realistically visit, as all food pantries within 20 miles of the customer’s home, a distance that captures 90% of observed pantry trips.¹² Because many pantries create geographic restrictions to limit their service area, I exclude pantries that are not available to a customer, such as pantries with zip code restrictions that would make the customer ineligible. Characterizing the choice set allows me to compare how access changes with a customer’s home location, since the pantries within reach directly determines the amount of charitable food a customer can receive. On average, households choose from just over 25 food pantries.

Table 2 presents descriptive statistics on each of the census-tract neighborhoods that food pantry customers live in by income quintile. Panel A includes the total number of customers living in a neighborhood that matches each income quintile and the total number of census tracts that fall into each income quintile. More than five times as many food pantry customers live in the lowest income

¹²If a customer visits a pantry beyond this radius, I include the selected pantry in the choice set, which extends the maximum distance to about 23 miles in practice.

neighborhoods compared to the number of customers living in the highest income neighborhoods. For the most part, the number of customers in each income quintile decreases with neighborhood income, suggesting a higher need in low-income neighborhoods.

Table 2: Demographics by Income Quintile

	Income quintile of customer's census tract				
	1	2	3	4	5
<i>Panel A. Customer counts by quintile</i>					
No. of customers	36,660	21,400	21,639	15,593	7,111
No. of census tracts	65	66	65	65	64
<i>Panel B. Census tract-level characteristics</i>					
Median income (\$)	40,936 (7,127)	56,297 (3,281)	68,532 (4,377)	84,767 (5,941)	136,924 (34,743)
Population	3,046 (1,125)	3,820 (1,005)	4,033 (1,159)	4,049 (1,577)	4,986 (2,032)
% With public benefits	0.12 (0.05)	0.06 (0.02)	0.04 (0.02)	0.03 (0.01)	0.01 (0.01)
% With bachelor's	0.06 (0.03)	0.08 (0.04)	0.10 (0.04)	0.13 (0.05)	0.22 (0.05)
% White	0.92 (0.37)	0.95 (0.04)	0.96 (0.16)	0.93 (0.18)	0.87 (0.13)

Notes: This table presents the count of food pantry customers in each income quintile as well as census-tract level descriptive statistics. Income quintiles are defined at the census tract level, relative to all census tracts within the 19 rural counties in the MOFC region. Row (1) of Panel A is the total of number of households living in each income quintile that visited a food pantry in the data, and Row (2) of Panel A is the total number of census tracts that fall into each income quintile. Panel B presents descriptive statistics at the census-tract level. Row (1) of Panel B gives the median income of the quintile at the census-tract level, not the income of the customer. Panel B also shows demographic details about the census tracts in each income quintile including the average population, the share of individuals on any form of public benefits, the share of individuals with a bachelor's degree, and the share of individuals who identify as white. (Source: author's calculations using MOFC and ACS data.)

Panel B of Table 2 gives demographic details of the neighborhoods in each income quintile where food pantry customers reside. The mean of the tract-level median income of the highest income neighborhoods is nearly \$100,000 more per year than the lowest income neighborhoods. Discussions with fundraisers for food pantries suggest that typical donors have a household income of \$75,000 or more, and so higher-income neighborhoods likely have more potential donors. Similarly, charitable giving and volunteerism increase with education level (Jones 2023). Table 2 shows that the fraction of households with a bachelor's degree increases with income quintile, again suggesting that there are more potential donors living in higher-income neighborhoods. Therefore, the differences in income

and education level may translate into higher pantry capacity in higher income quintiles.

On the other hand, according to MOFC's eligibility forms, a family of three earning 200% of the federal poverty line would earn \$53,300 annually. Therefore many households in the first and second income quintile neighborhoods are likely eligible for food pantry services, even if they are not visiting the pantry. Likewise, the fraction of households in each income quintile on public benefits, including SNAP or Temporary Assistance to Needy Families (TANF), decreases with neighborhood income quintile, again indicating that households in the lower income quintiles are more likely to be eligible for food pantry services in conjunction with other services. Altogether, the variation in high versus low income quintile demographics points toward potential misalignment between pantry resources and neighborhood demand.

Although census tract populations are generally similar, tracts in the lowest income quintiles have fewer residents, which likely reflects their rural location. Figure 2 shows the MOFC territory divided into census tracts. The darker blue shaded areas represent the higher income quintile neighborhoods while the lighter blue represent lower income quintile neighborhoods. The map shows that the higher-income neighborhoods tend to be clustered around Franklin County, while the lower-income neighborhoods are farther away toward the West Virginia border in Appalachian Ohio. The concentration of food pantries, which are represented by triangles, is much higher in Franklin County than in the rural counties, and there are fewer food pantries in the Appalachian region than there are near Franklin County. Some households live near as many as 130 pantries, while others only have two. Physical access constrains food pantry availability in rural areas, and therefore the few local pantries directly determine households' access to charitable food. If these pantries do not receive ample support, rural customers may face lower per-visit benefits.

The number of food pantries within a customer's choice set dictates whether customers can choose between different pantries. But each pantry is unique and offers a different level of service based on the pantry type and financial support. Table 3 demonstrates the variation in the amount of food distributed at the pantry-year level by pantry type. The table displays the average, annual amount of food ordered by each type of pantry. Approximately half of the food pantries are churches, with over 100 church pantries available each year throughout the sample time frame. Nonprofits represent the next largest type of pantry, with nearly 80 nonprofits each year. Finally, branch organizations represent about 25 organizations in a given year. In total, there are over 200 food pantries each year, with some entry and exit, mostly around the COVID pandemic in 2020 and 2021.¹³

While church food pantries are the most common, they distribute less food annually on average than nonprofits or branches. Churches order 88,910 lbs per year per pantry, compared to 272,201

¹³Over the six-year period from 2018 to 2023, 36 food pantries entered the sample. Of these, 22 entered during 2020 or 2021. A total of 19 pantries exited the sample, 12 of which were in 2021 or 2022. Appendix Figure 3 shows the total entry and exit by year.

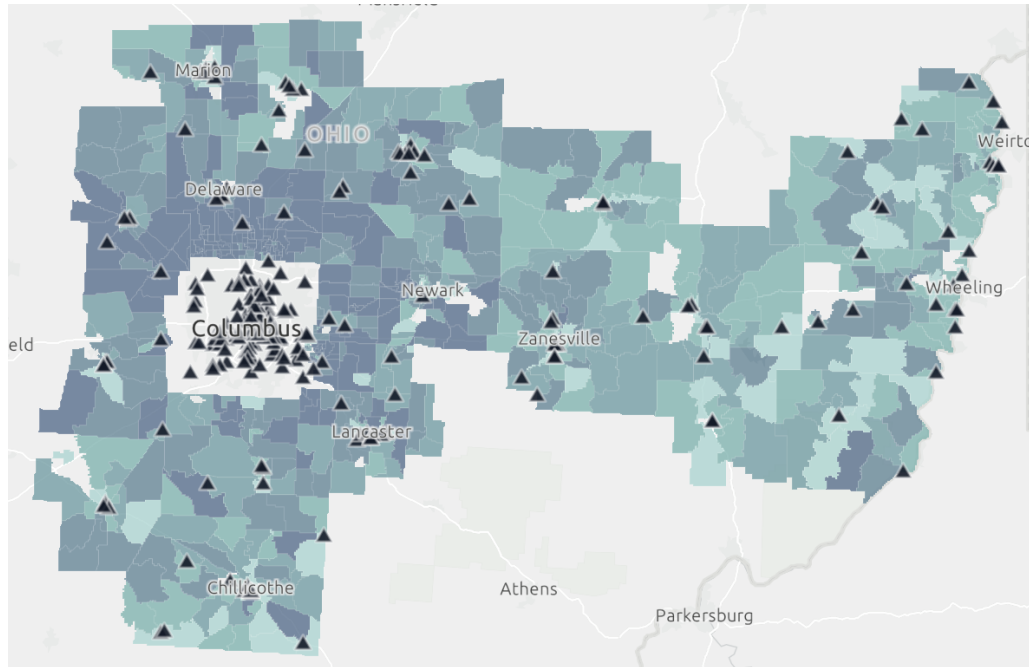


Figure 2: Map of Income Quintiles and Pantry Locations

Notes: Each polygon in the map represents a rural census tract. The darker colored polygons represent higher income quintiles, while the lighter colors represent lower income quintiles. The triangular points represent food pantry locations.

lbs per year per nonprofit pantry, which directly affects both the number of customers served and the amount of food each customer receives. Branches generally order the most food, though this average is heavily influenced by two large Mid-Ohio Markets that serve hundreds of customers daily. Excluding these two locations, branches distribute less food per year than nonprofits, averaging 151,664 lbs.¹⁴

Even within nonprofits, the amount of food distributed varies considerably by organization size and filing type; Appendix Table 2 illustrates this variation. The types of food each pantry orders also shape the service they provide. Overall, the distribution of food categories is similar across organization types, but churches tend to order a larger share of canned goods and a smaller share of fresh products, which may influence the nutritional content and overall quality of food available to customers.

Finally, Table 4 breaks down annual Form 990 data to highlight the finances of the largest nonprofit organizations analyzed in this paper. Though Form 990 filers are considered ‘large’ in the data, compared to national organizations, all nonprofit food pantries are small, with annual total revenue of just under \$2 million.¹⁵ The table shows considerable differences in the amount of funding each organization receives, which translates into differences in expenditures and community investment.

¹⁴Appendix Table 1 shows a breakdown of food ordered by branches excluding MOFC.

¹⁵For comparison, the American Red Cross earned over \$3 billion in total revenue in 2022, according to their Form 990 tax document.

Table 3: Annual Food Pantry Orders by Pantry Type

	Type of pantry			
	All	Church	Branch	Nonprofit
<i>Panel A. Total ordered</i>				
Quantity (lbs)	200,518 (676,718)	88,910 (115,689)	492,184 (1,711,087)	272,201 (448,499)
Price (\$)	8,569 (17,628)	4,790 (74,212)	18,805 (34,916)	10,775 (17,738)
<i>Panel B. Quantity per food category (%)</i>				
Bread	3.6 (6.0)	2.5 (4.5)	3.6 (5.8)	5.3 (7.6)
Shelf-stable	28.4 (16.9)	31.6 (17.1)	28.9 (16.8)	23.2 (15.3)
Fresh foods	43.3 (22.0)	39.4 (20.4)	46.1 (24.5)	48.4 (22.3)
Dairy	5.8 (4.4)	5.3 (4.1)	6.3 (5.0)	6.4 (4.6)
Meat	13.3 (8.4)	13.6 (8.5)	12.1 (8.4)	13.2 (5.1)
Snacks	11.6 (6.7)	13.0 (6.5)	9.9 (6.8)	10.1 (6.5)
Non-dairy beverages	7.4 (6.8)	8.1 (7.0)	7.4 (7.5)	6.4 (6.1)
Non-food necessities	4.5 (5.9)	4.1 (5.5)	4.1 (4.7)	5.1 (6.8)
Observations	1,354	727	170	457

Notes: This table shows the total amount of food in pounds that each type of food pantry orders on average, the amount spent on food on average, and the percent of food that falls into each category. Although most categories are self-explanatory, ‘Shelf-stable’ represents canned, boxed, or frozen foods, and ‘Non-Food Necessities’ includes items like cleaning products, personal hygiene products, diapers, and vitamins. The total price is the average amount spent on food and food delivery during each year for the pantries that ordered food. Note: fresh foods include meat, dairy, and produce. The food bank offers fresh food for free to all pantries. (Source: author’s calculations using MOFC data.)

Table 4: Annual Finances of Nonprofit Pantries

	Nonprofit	990	990-EZ
Total Revenue (\$1,000s)	1,942 (3,742)	2,542 (4,132)	86 (50)
Government Grants (\$1,000s)		1,598 (3,130)	
Total Contributions (\$1,000s)	1,578 (2,657)	2,062 (2,896)	78 (47)
Program Revenue (\$1,000s)	371 (1,897)	429 (2,040)	6 (18)
Total Expenditures (\$1,000s)	1,838 (3,557)	2,406 (3,928)	80 (49)
Rent & Utilities (\$1,000s)	17 (62)	21 (70)	7 (13)
Total Salary Expenditure (\$1,000s)	668 (1,667)	880 (1,870)	9 (15)
Employees		34 (65)	
Volunteers		634 (1,383)	
Observations	213	161	52

Notes: This table presents summary statistics about the nonprofit organizations that file a tax document, therefore excluding Form 990-N and nonfilers. All nonprofit filers in the sample are in the first column, nonprofits that file a Form 990 in the second column, and nonprofits that file a Form 990-EZ in the third. The Form 990-EZ does not contain information on government grants, employees, or volunteers, so these variables are omitted in the first and third columns. Government Grants is conditional on receiving a government grant, for a total of 79 pantry-year observations. (Source: author's calculations using NCCS and IRS data.)

A majority of the annual revenue for food pantries comes from contributions from donors, corporations, and private foundations. Large nonprofits, or those that file a Form 990, receive over two million dollars in contributions, while mid-sized nonprofits, those that file a Form 990-EZ, bring in closer to \$80,000. Government grants are only reported for large nonprofits and the average grant amount was nearly \$1.6 million.¹⁶ A much smaller portion of an organization's budget comes from program revenue, which includes money generated from activities related to the nonprofit organization's mission. For instance, a community health nonprofit might charge fees for cooking classes or sell boxes of community-supported agriculture (CSA) from a farm it operates. These activities

¹⁶Only the Form 990 breaks contributions into 'Government Grants' and 'Total Contributions' on the Form 990 and not on the Form 990-EZ. However, as shown in the Table 4, government grants tend to be large, meaning that organizations that receive them likely do not fall under the threshold to file a Form 990-EZ. Large nonprofits received a government grant in 79 of the 161 pantry-year observations.

generate revenue, but the proceeds are typically reinvested into the nonprofit rather than distributed as profit. The total revenue from programs for mid-sized nonprofits is \$5,800, while program revenue for large nonprofits is \$429,400, larger than the total revenue earned by mid-sized nonprofits.

I also include details on salary expenditure and rent and utilities. Larger organizations spend more on these expenses, likely because they have more staff and operate more programs than the mid-sized nonprofits. Larger nonprofits that operate a food pantry are more likely to offer other social services, such as temporary housing, daycare, and senior transportation. These organizations employ a full staff which may include social workers, teachers, fundraisers, and program directors, and therefore not all staff supports the food pantry. Additionally, managing large public funds, such as the government grants detailed in Table 4, requires administrative personnel dedicated to regulatory compliance, data tracking, and financial auditing, which cannot be managed by volunteer labor. Additionally, these large organizations almost always maintain at least one physical space that they own or rent.

Most mid-sized nonprofits, on the other hand, typically operate a food pantry only, and therefore survive with a smaller staff, or as an all volunteer organization. Mid-sized nonprofits spent some funds on rent and utilities in 18 pantry-year observations, and spent some funds on salaries in 15 pantry-year observations. Organizations that do not use funding for staff rely entirely on volunteers, while organizations that do not spend on rent and utilities could be relying on donated space. Differences in total revenue, staffing, and rent expenditures across pantries translate into differences in the amount and type of food they can distribute. These operational constraints help explain variation in household per-visit benefits across neighborhoods.

4 Conceptual Framework

My primary objective is to estimate how local economic conditions leads to disparities in the level of service that food pantries can provide in different geographic areas and, in turn, how variation in the quality of food pantry service contribute to disparities in consumer welfare. Food pantries share a common goal of reducing hunger and each food pantry focuses on fighting hunger in their own community subject to their own funding constraints. This system likely results in fewer individuals facing hunger in communities with food pantries, but does the community-centered approach lead to the greatest aggregate benefit for food pantry customers?

In this section, I consider how pantries determine their service design and food distribution levels based on the amount of funding and community support they receive. Formally, I indicate the amount of funding as $f(d, v)$, representing the total amount of pantry resources available to alleviate hunger for a pantry in the community. Community members contribute money (d) and volunteer time (v) to support the pantry's mission.

Each customer i derives utility, $u_i(b, \mathbf{x})$, from visiting a pantry. Utility depends not only on the

amount of food received from the pantry, b , but also on the pantry's characteristics, represented by the vector $\mathbf{x}$. Pantry characteristics includes the total hours of operation in a month, the specific days (e.g. Monday, Sunday) and times (e.g. mornings, evenings) the pantry is open, limits on the number of visits allowed per month, and pantry type (e.g. church, nonprofit). For example, a customer may need to take off work to visit a pantry that is only open on Monday mornings, and may find visiting a pantry open on Friday evenings more convenient. Service design also includes the types of food that the pantry offers. A customer may value a bundle of products that includes dairy, meat, and produce offered at one pantry more than a bundle of products that includes canned vegetables, rice, and juice offered at another pantry, even if the total amount of food is the same.

Although the food a customer receives from the pantry is free to the customer, the customer does incur some cost with each visit, $c(\text{dist}_i, \mathbf{z}_i)$. The cost is a function of the distance that customer i travels to the pantry, dist_i . A customer's sensitivity to travel costs varies with the customer's characteristics, like household size, neighborhood income, and transportation infrastructure, including public transportation and road quality. These customer characteristics are represented by $\mathbf{z}_i$.

Distributing food is costly to the food pantries, and the amount of funding received by each pantry serves as a budget constraint. A pantry's cost is a function of the total amount of food purchased to distribute, $B = \sum_i b_i$, as well as the service design they choose, $\mathbf{x}$. Each pantry will choose the amount of food to distribute in order to maximize the benefit that the customers who visit the pantry gain from each visit constrained by f , such that $f(d, v) = c(B, \mathbf{x})$. The subset of customers who visit a given pantry is represented by $I(b, \mathbf{x})$ and is a function of the food pantry's characteristics and amount of food distributed.

Formally, the pantry solves the following problem:

$$\begin{aligned} \max_b \quad & \sum_{i \in I(b, \mathbf{x})} u_i(b, \mathbf{x}) - \sum_{i \in I(b, \mathbf{x})} c_i(\text{dist}_i, \mathbf{z}_i) - c(B, \mathbf{x}) \\ \text{s.t.} \quad & c(B, \mathbf{x}) = f(d, v) \end{aligned} \tag{1}$$

Food pantries each face an individual budget constraint based on the support from their own donors and volunteers, where donors and volunteers give to pantries according to their own preferences. Volunteers, for instance, may choose to work at a pantry that is easy for them to drive to and is located in a neighborhood where they are comfortable. Donors may give to a pantry that they are familiar with, perhaps because they see it on their way to work or because it is promoted at their place of worship. In other words, it is likely that the amount of funding each pantry receives is not solely driven by demand for the food pantry's services, rather by donor and volunteer preferences. This means that aggregate welfare could possibly be improved by redistributing the total amount of funding received by food pantries according to *customer* demand, rather than donor preferences. In other words, there

may be alternative distributions of the total amount of funding received by pantries that improves aggregate welfare.¹⁷

While pantries aim to maximize the benefits provided to their customers, they face limited ability to adjust their service design. In practice, changes to pantry operations, like extending hours of operation, upgrading refrigeration systems, or modifying physical locations operate as fixed-cost tied to long-run investments. As a result, pantries cannot optimize on service characteristics, and therefore x represents a fixed constraint in the pantry's problem. Pantries can, however, adjust the amount of food they distribute, (B). Conditional on this binding food funding constraint, cross-pantry variation in fixed service characteristics (x) can be treated as plausibly exogenous to unobserved, customer preferences.

To understand how local economic conditions and donations may result in a distribution of funding that restricts access in areas that need the most support, consider two food pantries. The first pantry is in a low-income area with many potential customers and few donors and volunteers, and therefore the pantry is resource-constrained. This restrictive budget and high demand means that the pantry will limit the amount of food they give away so that they are able to meet their budget restriction.¹⁸ In this stylized example, customers who visit the pantry will receive less utility per visit than they would have if the pantry received more funding. If these households are particularly income-constrained, this may mean they are unable to adequately meet their food needs.

The other pantry is in a higher-income area and not only receives significant contributions from donors and volunteers, but also has very few customers. This means that the customers that visit the pantry in the high-income area receive a lot of utility from each visit. Additionally, because the pantry in the high-income area has more resources, customers from the low-income area may travel farther to access the higher-capacity pantry. This would suggest that the high-capacity pantry provides so much utility that visiting the pantry is worth the additional travel costs for customers in low-income areas. Because households experience diminishing marginal utility with respect to the amount of food received during each visit, reallocating funding from the high-capacity pantry in the high-income area to the low-capacity pantry in the low-income area could increase utility for the low-income customers without meaningfully decreasing utility for the high-income customers. Customers in the low-income area may not need to travel as far to receive high-quality pantry service, for example, or they may be

¹⁷Improving aggregate welfare also assumes that the utility of donors and volunteers increases if aggregate customer utility is maximized. However, if supporters prefer to increase utility only for customers within a certain region, for instance, if they prefer to support their neighbors, it is possible that donor and volunteer utility would decrease if funding were reallocated.

¹⁸In practice, meeting the lower budget constraint may involve limiting the number of visits each customer is allowed in a given month. Alternatively, some pantries are open for just a few hours in a month, restricting their service to only the customers who are able to visit the pantry during those hours. This often means that pantries are open during inconvenient hours, such as during the workday. Some pantries will offer food distribution on just a few days a month, and will distribute food until it is all gone, meaning that individuals run the risk of waiting for food that they ultimately will not receive.

able to visit the pantry more frequently because the pantry can now offer more frequent distribution.

In the next sections, I analyze the geographic distribution of food pantry characteristics to determine if pantries in low-income areas impose more restrictions on their customers. I then estimate a demand model to quantify how variation in food pantry access impacts the value derived from each food pantry visit. Using the parameters from the demand model, I simulate an alternative funding distribution that improves aggregate consumer welfare by increasing the utility of customers in low-income areas without substantially decreasing the utility of customers in high-income areas.

5 Descriptive Analysis

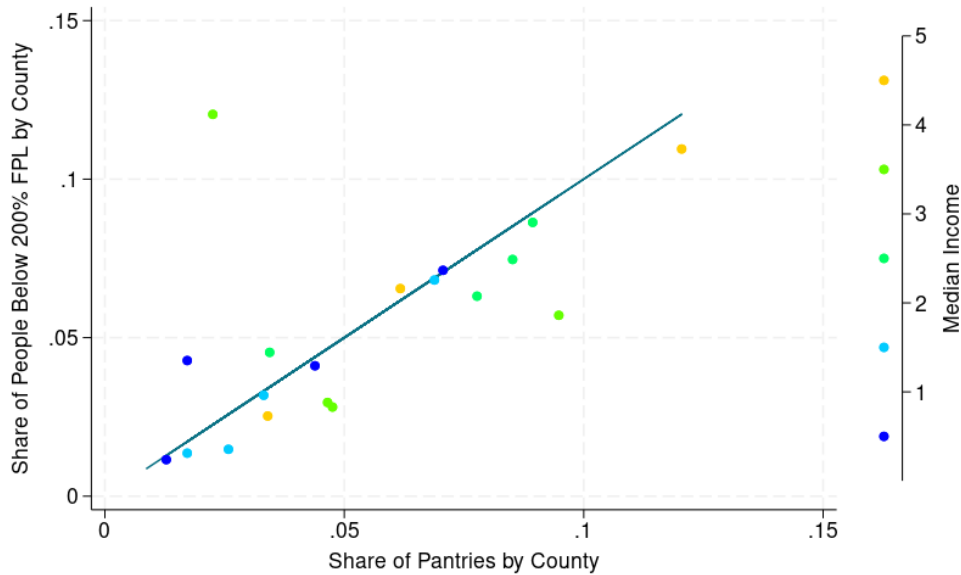
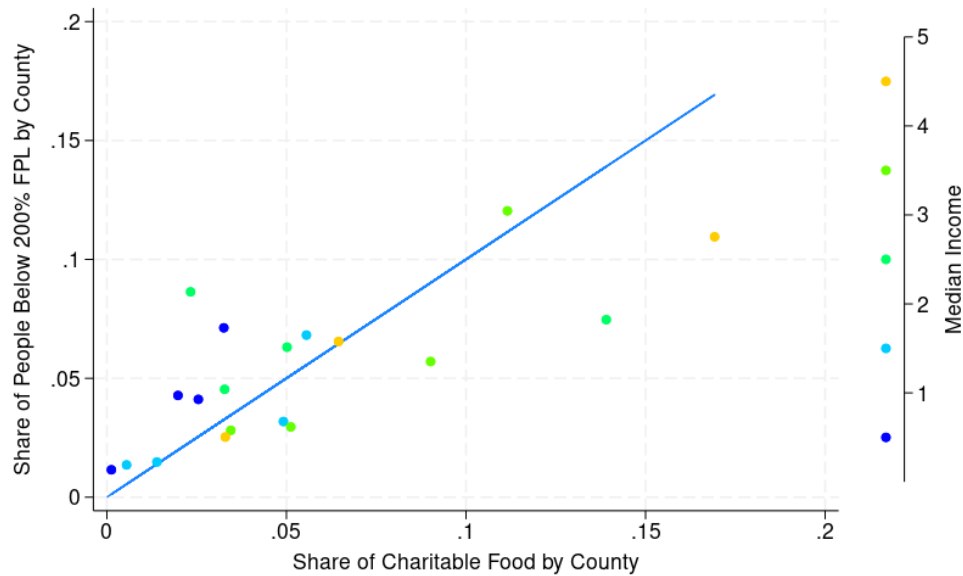


Figure 3: Share of Pantries vs. Share of Pantry-Eligible Individuals by County

Notes: The figure plots the share of households below the 200% of the federal poverty line in each county against the share of food pantries in each county, and a 45 degree line. Points above the line have a disproportionately high share of individuals below 200% of the federal poverty line relative to the share of food pantries, while points below the line have a disproportionately high share of pantries relative to the number of eligible individuals. Each point is colored according to the county income quintile with yellow points representing the highest income quintile counties and dark blue points representing the lowest income quintile counties.

To determine whether the data are consistent with the hypothesis that households in higher-income areas live near more, well-resourced food pantries compared to customers in lower-income areas, I begin by broadly considering the number of food pantries within a given county compared to the number of people eligible for services. If pantry locations were driven by customer travel costs, the distribution of pantries would align with the share of eligible customers living in each county; for



above the 45-degree line, meaning that these low-income communities receive a substantially smaller share of the region's food than the population warrants. Conversely, almost all higher-income counties are located below the line, suggesting that they distribute a disproportionately high share of food resources to their communities. Together, these figures illustrate the underlying spatial mismatch: while low-income counties may have an adequate number of physical pantry locations, those specific pantries are under-resourced compared to their high-income counterparts.

Next, Table 5 shows how individual food pantry access varies for customers residing in different income quintiles, indicating that food pantry customers living in the highest income quintiles have access to more pantries of every type within their choice set. Panel A shows that there are nearly 80 food pantries within 20 miles of a customer living in the highest-income neighborhoods and only 17 pantries within 20 miles of a customer living in the lowest-income neighborhoods. Customers living in high-income neighborhoods also live near more pantries with more flexibility, as shown in Panels B-D. On average, customers in high-income areas live near 46 pantries that are open at least weekly, 52 pantries open on weekends, and eight pantries that do not restrict the number of visits a customer can make. Customers in the second-lowest income quintile neighborhoods live near fewer pantries than those in the lowest-income quintile neighborhoods, with only 14 food pantries nearby and only seven open at least weekly.

The descriptive statistics in Table 5 support the theory that pantries in low-income areas are more restrictive than pantries in high-income areas on average. However, recall that the highest-income neighborhoods are more populous and tend to be clustered around Franklin County, suggesting that the apparent advantage in food pantry access may reflect easier access to urban spaces. To test whether the variation in pantry access reflects differential access to financial support and other resources, I look at how the number of nearby pantries varies with neighborhood income controlling for population and proximity to Franklin County using the following specification:

$$Y_i = \alpha + \sum_{q=2}^5 \beta_q \cdot \mathbf{1}[\text{Income Quintile}_i = q] + \beta_6 \text{Pop}_i + \beta_7 \mathbf{1}[FC_i] + \varepsilon_i \quad (2)$$

where Y_i is the number of pantries in individual i 's choice set and β_q represents the change in Y_i associated with a change in individual i 's census-tract-level income quintile, relative to the lowest income quintile. Pop_i is a control for the population of individual i 's census tract and $\mathbf{1}[FC_i]$ is an indicator equal to one if individual i lives in a county adjacent to Franklin County. β_q captures the difference in the number of food pantries within the choice set of a household living in each income quintile, relative to income quintile one. I plot the β_q coefficients and 95 percent confidence intervals in Figure 5 below.

The results reveal disparities in pantry access even when controlling for population and proximity

Table 5: Pantry Access by Income Quintile of Customer's Census Tract

	Income quintile of customer's census tract				
	1	2	3	4	5
<i>Panel A. Number of nearby pantries</i>					
Total	16.7 (16.4)	14.1 (12.2)	22.7 (23.7)	49.8 (40.2)	78.5 (43.3)
Nonprofits	5.6 (5.1)	5.0 (5.0)	7.9 (7.9)	16.2 (16.2)	24.9 (12.6)
990	1.5 (2.6)	1.3 (2.2)	2.6 (3.9)	6.3 (6.6)	9.9 (8.0)
990-EZ	0.7 (0.8)	0.6 (0.6)	0.8 (1.0)	1.7 (1.3)	2.5 (1.3)
Postcard	1.2 (1.3)	1.2 (1.3)	1.8 (1.6)	2.5 (1.8)	3.0 (1.9)
Churches	9.2 (9.4)	7.5 (7.1)	11.8 (13.4)	26.6 (22.8)	43.1 (25.3)
Branch	1.9 (2.3)	1.5 (1.9)	2.8 (3.1)	6.6 (5.5)	9.8 (5.5)
<i>Panel B. Open frequency</i>					
Open Weekly	14.1 (14.2)	7.3 (8.0)	12.5 (14.1)	29.2 (24.0)	45.9 (25.3)
Open Semi-Monthly	3.4 (3.6)	3.2 (2.4)	4.7 (5.3)	10.0 (9.2)	16.9 (10.7)
Open Monthly	3.2 (2.7)	2.7 (2.2)	3.9 (3.3)	7.2 (4.7)	10.2 (5.1)
Infrequent	1.1 (1.1)	0.9 (0.9)	1.4 (1.7)	2.9 (2.8)	4.9 (3.1)
<i>Panel C. Weekends</i>					
Open on weekends	11.0 (11.0)	9.3 (8.6)	15.0 (15.7)	32.9 (26.0)	51.5 (28.0)
<i>Panel D. Visit restrictions</i>					
Weekly Cap	14.1 (14.2)	11.9 (10.5)	19.1 (20.6)	42.2 (34.5)	66.7 (37.4)
Monthly Cap	1.2 (1.1)	0.9 (0.9)	1.2 (1.0)	2.0 (1.5)	2.8 (1.5)
No Cap	1.5 (1.9)	1.4 (1.8)	2.4 (2.7)	5.2 (4.9)	8.4 (5.4)
Observations	36,660	21,400	21,639	15,593	7,111

Notes: This table gives a count of the average number of food pantries within 20 miles of a customer living within each income quintile by pantry characteristics. Panel A breaks down the number of pantries by pantry type, including church, branch, and nonprofit. Panel B gives the average number of pantries that are open weekly, semi-monthly (2+ weeks per month), monthly, or less than monthly. Panel C gives the count of pantries open on Saturday or Sunday. Panel D presents the count of pantries that restrict the number of visits each customer is allowed to weekly or monthly, along with pantries that do not restrict visits. (Source: author's calculations using MOFC data.)

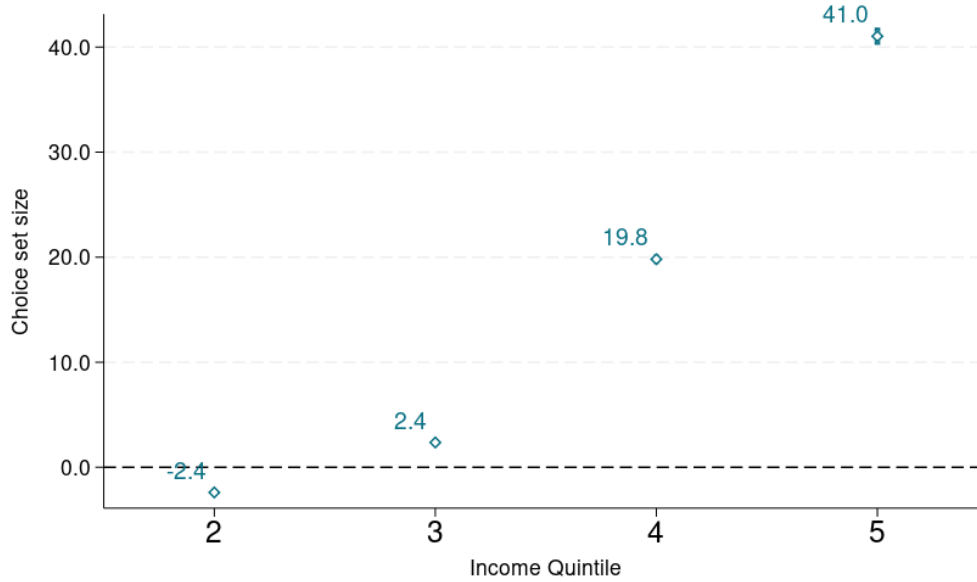


Figure 5: Variation in Total Options in Choice Set by Income Group

Notes: This figure plots the β_1 coefficients and 95 percent confidence intervals from Equation 2, which gives the change in the count of food pantries in the choice set of a customer living in each income quintile, relative to the count of pantries in the choice set of a household living in lowest income quintile neighborhood.

to Franklin County. Customers in the highest income quintiles have 41 more food pantries in their choice set relative to households in the lowest income quintiles. Additionally, customers in the highest-income neighborhoods have more pantries of every type near their homes compared to households in the lowest-income neighborhoods, including 13 more nonprofit food pantries, five more pantries without visit restrictions, and 27 more pantries open on weekends.¹⁹

Finally, Table 6 considers how neighborhood income relates to a lack of access to pantries with certain characteristics. Just over 3% of food pantry customers in low-income neighborhoods do not live near a nonprofit food pantry, for a total of 1,143 customers without access. Given that nonprofit food pantries distribute significantly more food than other types of food pantries, these customers may receive less food overall than customers in higher income areas. On the other hand, all customers in the highest two income quintiles have access to at least one nonprofit food pantry.

One of the most striking disparities is the lack of access to ‘flexible’ pantries. While only 4% of customers in high-income neighborhoods lack access to a pantry with unlimited visits, nearly 28% of

¹⁹I test the sensitivity of this result to different choice set radii in Appendix B. The tests show that the number of food pantries within a fifteen-mile radius increases with income quintile, similar to the twenty-mile radius. Within a ten-mile radius, households in highest-income neighborhoods have three more food pantries than those in the lowest income quintiles. Within a five-mile radius, households in the lowest income quintile have just three more food pantries in their choice set compared to households in the highest income quintiles, even though the number of customers in the lowest income group is five times larger than the number of customers in the highest income group.

customers in the lowest-income neighborhoods (over 10,000 individuals) are restricted to pantries with visit limits. As these limits are typically supply-driven—instituted when demand outstrips available food—this gap provides direct evidence of the resource constraints facing organizations in high-need areas.

Table 6: Percent of Customers Lacking Nearby Pantries with Specific Characteristics (by Tract Income Quintile)

	Income quintile of customer's census tract				
	1	2	3	4	5
<i>Panel A. Type of pantry</i>					
Nonprofit	3.12	1.25	2.63	0.00	0.00
Church	0.00	0.00	0.02	0.03	0.80
Branch	16.00	14.16	14.00	5.49	1.66
<i>Panel B. Open frequency</i>					
Open weekly	0.23	0.31	0.20	0.00	0.00
<i>Panel C. Weekend availability</i>					
Open on weekends	1.01	0.64	0.51	0.00	0.00
<i>Panel D. Visit restrictions</i>					
Weekly Cap	0	0	0	0	0
No Cap	27.74	34.80	16.90	11.31	4.06

Notes: Table 6 presents the percent of food pantry customers living in each income quintile that do not have a pantry within twenty miles of their residence with certain characteristics. Panel A breaks down the percent of customers without access by pantry type, and Panel B shows the percent that do not have access to a pantry that is open weekly. Panel C shows the percent of people that do not have a pantry open on Saturday or Sunday. Panel D shows the percent of people without access to a pantry that allows weekly visits or does not restrict visit frequency. (Source: author's calculations using MOFC data.)

Altogether, these findings emphasize a spatial mismatch within the charitable food network. While food pantry customers are more likely to live in low-income neighborhoods, generating the highest demand for food assistance, the pantries near these communities face capacity constraints resulting in restricted hours, limited food volume, and visit caps. In contrast, households in higher-income areas live near more flexible and well-resourced pantries, despite representing a smaller share of the eligible population. This evidence suggests that the current food pantry funding model restricts access for the highest-need populations, however it cannot quantify actual welfare losses. To measure exactly how these capacity constraints impact consumer utility, and to evaluate whether an alternative funding allocation could improve welfare, I estimate a structural model of food pantry demand.

6 Demand Estimation

To quantify how variation in food pantry access translates to differences in the welfare gained from visiting a pantry, I estimate a discrete choice model that allows for heterogeneous consumer preferences. Each week, customers make a discrete choice between visiting a food pantry within a 20-mile radius of their home location²⁰ or the outside option of sourcing food from food retailers. Formally, let $\mathcal{J}_i$ denote the set of food pantries located within 20 miles of customer i 's home location. Then customer i faces a discrete choice between these nearby pantries (indexed by $j \geq 1$), or the “outside option” (indexed by $j = 0$). The customer, then, will choose the option that maximizes her utility.

Customer i 's conditional indirect utility u for choosing pantry j during week t is represented by:

$$u_{ijt} = \mathbf{x}_{jt}\boldsymbol{\beta}_i - \alpha_i D_{ij} + \varepsilon_{ijt}, \quad (3)$$

where $\mathbf{x}_{jt}$ is a vector of pantry characteristics, including pantry type (nonprofit, church, or branch), visit restrictions, weekly opening times, and the quantity of food in different categories (e.g., produce and dairy) that a customer can expect to receive per visit. Because households do not observe current pantry inventories prior to visiting, I construct expected food availability using annual averages. Specifically, I divide each pantry's total annual food distribution by its total number of annual visits. This measure captures persistent differences in pantry capacity and resources across locations, while smoothing over short-run fluctuations in supply that are unlikely to be observed by households. This approach isolates systematic differences in expected resources across pantries.

To capture variation in the attractiveness of the outside option, I include the density of SNAP-authorized retailers within a five-mile radius of a customer's home.²¹ This allows the model to account for the fact that a customer's likelihood of visiting a pantry depends on the availability of local food retailers; a higher concentration of nearby food retailers makes it easier for customers to substitute between a food pantry and a food retailer.²²

The customer's utility decreases in the Euclidean distance D_{ij} ²³ from the customer's home to the food pantry. In Equation (3), the coefficient α_i measures customer i 's sensitivity to distance.²⁴

²⁰The choice set is adjusted to account for a food pantry's open times such that a customer is only choosing among the pantries open at least one day in week t . For example, if a pantry is only open the first week of the month, that pantry is only included in a customer's choice set during the first week of the month.

²¹To identify nearby food retailers, I use data from the USDA's Historical SNAP Retailer Locations (U.S. Department of Agriculture, Food and Nutrition Service 2025a). I also include the count of SNAP-retailers within a five-mile radius of the customer's home location, which includes the latitude and longitude of retailers that accept SNAP benefits. I match each customer's home location with all food retailers within a five-mile radius of their home location. Retailers that accept SNAP must sell or stock a certain amount of staple food items, including cereals, dairy, meat, fruits and vegetables. Using SNAP retailers excludes some retail options, such as restaurants or grocery retailers that do not meet SNAP requirements.

²²Appendix Table 3 details differences in household size and food retailers by income quintile.

²³Using the Euclidean distance rather than travel distance is an imperfect measure of the costs of visiting the food pantry, likely underestimating the true cost of a visit.

²⁴Other costs to food pantry service may include the time spent waiting for food at the pantry, childcare costs, foregone

Because food pantries offer food free to their customers, travel time is the only observable direct cost that customers face. Finally, customer i exhibits idiosyncratic likes or dislikes ε_{ijt} for pantries j at time t , which I assume are independently and identically distributed according to a Gumbel distribution.

I model customer i 's preferences as a function of household-level demographics H_{it} , including household size and neighborhood income quintile. Including household size captures how preferences vary with different levels of a household's need for food; customers in larger households may prefer pantries with different characteristics than customers in smaller households, for example. Finally, allowing preferences to vary by income quintile tests whether customers living in different income quintile neighborhoods experience different sensitivities to travel or different preferences for food pantry characteristics.

Letting h index various household demographics, distance sensitivity is specified as:

$$\alpha_i = \alpha + \sum_h \alpha_h H_{it}^h. \quad (4)$$

In addition, I allow customers' tastes for pantry characteristics to vary by customer demographics as well. Specifically, a customer's taste β_{ik} for pantry characteristic k is given by:

$$\beta_{ik} = \beta_k + \sum_h \beta_{kh} H_{it}^h. \quad (5)$$

This model identifies how variation in food pantry service design and expected food availability affects the utility that customers from different neighborhoods derive from each pantry. I assume that the idiosyncratic preference term, ε_{ijt} , is uncorrelated with the explanatory variables in the utility function. In particular, short-run fluctuations in pantry resources are assumed not to alter the observable service characteristics captured by $\mathbf{x}$. As discussed in Section 4, pantries face substantial constraints in adjusting operational features such as hours of operation or visitation policies. I further assume that temporary funding changes do not affect the demographic composition of the customers served by pantry j , denoted by $\mathcal{I}$. Instead, these fluctuations primarily influence the quantity of food available for distribution, while leaving the underlying structure of pantry services unchanged.

Another assumption is that the distance cost that customers face is the travel from their home location to the food pantry. It is possible that households visit a food pantry in a multi-stop trip, such as from work, also known as trip chaining. If households visit the pantry as a part of a trip chain, the distance calculated from a customer's home to the pantry misrepresents travel costs.

Evidence from grocery shopping literature indicates most households visit the grocery store directly from home, and finds that only 8% of grocery store trips are directly from work or part of a trip chain from work, 28% of grocery store trips are part of a trip chain to and from home, and 64% of

labor hours, or the stigma associated with visiting a food pantry.

grocery store visits are directly from home (Ver Ploeg et al. 2015). Food pantry visits are often by appointment or involve long wait times, making trip chaining less likely. Although my data do not include information on wait times, Gose (2023) found that customers visiting Atlanta food pantries waited between ten minutes to two hours to receive food. Conversations with food pantries in the MOFC network confirm that wait times can be lengthy for customers in the MOFC region as well, particularly when the pantry opens. Therefore, households that would like to pick up food from a food pantry probably need to schedule around the distribution times rather than their own schedules.

7 Results

Table 7 reports the estimated coefficients from a model without the demographic interactions to demonstrate how food pantry characteristics shape the level of utility derived from a food pantry visit on average.²⁵ Broadly, the estimates reveal patterns in food pantry preferences related to the (i) opportunity cost of visiting the food pantry, (ii) the food pantry’s service design, and (iii) the concentration of local retailers, or the outside option, that a customer can choose from.

The results indicate that households prefer pantries that are closer to home, as expected. Because travel distance represents the primary cost of visiting a pantry in this setting, this finding suggests that households seek to minimize travel costs, all else equal. Customers also favor pantries with distribution times in the mornings or evenings and on Fridays, while distribution times on other days have comparatively small effects on pantry choice. Translating these preferences into distance equivalents, customers would be willing to travel an additional 1.2 miles to visit a pantry open in the mornings or evenings, and an additional 1.7 miles to visit a pantry open on Fridays.²⁶

Compared to distribution times, the amount of food offered per visit does not affect utility significantly, as shown in Panel C of Table 7. Customers may slightly prefer pantries that offer more bread, dairy, juice or other beverages, and snacks per visitor over pantries that offer more shelf-stable food and non-food necessities, such as toothpaste, diapers, or detergent.²⁷ This suggests that for the average customer, the presence of a reliable food source and its accessibility are primary concerns, while the amount and types of food received are second-order. Panel D of Table 7 shows that customers also have a stronger preference for nonprofit pantries and would travel an additional 1.2 miles to visit a nonprofit pantry, all else equal.

Finally, Panel E of Table 7 shows how increasing the number of nearby food retailers affects the

²⁵A table of the base coefficients estimated from a model with the demographic interactions is available in Appendix Table 4.

²⁶I calculate these by taking the ratio of the characteristic coefficient to the distance coefficient, $(\frac{\beta}{\alpha})$, where α is the distance coefficient.

²⁷Snacks include items like pudding, desserts, and potato chips, but also items like salad dressing and pizza dough. Shelf-stable food includes canned fruits and vegetables, pasta, and frozen meals like pizza or burritos. Dairy includes milk, yogurt, cheese, butter, and eggs.

Table 7: Coefficient Estimates for Model without Demographic Interactions

	Point estimate	Standard error
<i>Panel A. Travel cost</i>		
Distance $\times -1$	0.259	0.000
<i>Panel B. Pantry flexibility</i>		
No cap on groceries obtained	−0.691	0.006
Limit one visit per week	−1.871	0.006
Open mornings	0.317	0.003
Open afternoons	0.073	0.003
Open evenings	0.313	0.003
Unknown open hours ^a	−0.780	0.006
Open Sundays	0.090	0.006
Open Mondays	−0.349	0.002
Open Tuesdays	−0.063	0.003
Open Wednesdays	0.153	0.003
Open Thursdays	−0.082	0.003
Open Fridays	0.429	0.003
Open Saturdays	−0.120	0.003
<i>Panel C. Food per customer</i>		
Bread per visitor (lb)	0.012	0.000
Shelf stable food per visitor (lb)	−0.024	0.000
Fresh food per visitor (lb)	−0.003	0.000
Dairy products per visitor (lb)	0.045	0.000
Meat per visitor (lb)	−0.008	0.000
Snacks per visitor (lb)	0.031	0.001
Non-dairy beverages per visitor (lb)	0.011	0.000
Non-food necessities per visitor (lb)	−0.025	0.000
Unknown no. of visits ^b	−0.926	0.003
<i>Panel D. Pantry type</i>		
Church	−0.056	0.004
Mobile pantry or produce market	−0.926	0.003
Nonprofit	0.306	0.003
<i>Panel E. Retail availability</i>		
Outside option $\times$ no. nearby grocers	−0.019	0.000
Outside option $\times$ no. nearby local food stores	0.091	0.002
Outside option $\times$ no. nearby small retailers	−0.003	0.000
Outside option $\times$ no. nearby supermarkets	0.114	0.001

Notes: This table reports coefficient estimates for a conditional logit model of food pantry choice with neighborhood income-invariant preferences (see Equation (3)). Concerning pantry type (Panel D), the omitted type of pantry is branch locations. The rightmost column reports heteroskedasticity-robust standard errors.

^a Open hours cannot be inferred because pantry does not report visit times.

^b When the number of weekly visits is unknown, this variable is defined as one and the per-customer food variables are set equal to zero.

value of the outside option. Customers who live near more supermarkets, defined as large retailers with more than ten checkout lanes that sell groceries and other merchandise, or local food stores, including specialty stores (e.g., bakeries, butchers, or produce stands), farmers markets, and food co-ops, derive more benefit from the outside option. One interpretation of this result is that ample nearby retailers give customers attractive alternatives to the food pantry, so choosing to visit the pantry despite the outside options suggests that each food pantry visit provides a high value to the customer. Alternatively, more supermarkets and local food stores may improve the pantry itself; for example, if a local bakery donates day-old pastries to their nearby food pantry, this may make the food pantry's offerings more valuable to the customer. By contrast, the concentration of small retailers (like dollar stores and convenience stores) does not meaningfully affect the value of the outside option. This suggests that households seldom substitute between food pantries and small retailers. The same seems to be true of grocery stores, defined as stores that primarily stock food rather than food and other merchandise.

Table 8 presents the estimates including the demographic interactions (the α_h 's and β_{kh} 's), and reveals how changing household characteristics shift preferences for pantry characteristics. The distance-related coefficients suggest that travel sensitivity increases with neighborhood income. Even among food pantry customers, households residing in higher-income neighborhoods may face higher opportunity costs of travel, making proximity more important.

Higher-income households also show stronger preferences for pantries that distribute in the afternoons or on Fridays, compared with lower-income households. This suggests that timing flexibility matters more for households with higher opportunity costs. Preferences for pantry type also vary by income quintile, as shown in Panel D of Table 8, which reveals that households in neighborhoods in the second or third income quintiles actually have a stronger preference for church pantries compared to households in the lowest income quintiles.

Table 8: Coefficient Estimates for Model with Demographic Interactions

Pantry characteristics	Demographic interactions (β_{kh} 's and α_h 's)					
	Base coefficients (β and α)	Income quintile of customer's census tract				Household size
		2	3	4	5	log(size) Unreported
Distance $\times -1$	<i>Panel A. Travel cost</i>					
	<i>Panel A. Travel cost</i>					
	0.223 0.001	0.018 [0.001]	0.023 [0.001]	0.053 [0.001]	0.056 [0.001]	0.008 [0.000] 0.497 [0.010]
<i>Panel B. Pantry flexibility</i>						
No cap on groceries obtained	-0.664 0.006					
Limit one visit per week	-1.800 0.006					0.018 [0.003] -3.325 [0.107]
Open mornings	0.294 0.003					
Open afternoons	-0.121 0.005	0.277 [0.006]	0.131 [0.006]	0.421 [0.007]	0.330 [0.010]	
Open evenings	0.288 0.003					
Unknown open hours	-0.707 0.007					-0.121 [0.004] 2.624 [0.006]
Open Sundays	0.068 0.006					
Open Mondays	-0.328 0.002					
Open Tuesdays	-0.072 0.003					
Open Wednesdays	0.167 0.003					
Open Thursdays	-0.062 0.003					
Open Fridays	0.320 0.004	0.094 [0.007]	0.209 [0.006]	0.150 [0.007]	0.198 [0.010]	
Open Saturdays	-0.090 0.004					-0.018 [0.004] -1.236 [0.127]

Table 8 (continued)

Pantry characteristics	Demographic interactions (β_{kh} 's and α_h 's)					
	Income quintile of customer's census tract			Household size		
	2	3	4	5	log(size)	Unreported
<i>Panel C. Food per customer</i>						
Bread per visitor (lb)	0.011 0.000					
Shelf stable food per visitor (lb)	-0.025 0.000				0.000 [0.000]	-0.436 [0.007]
Fresh produce per visitor (lb)	-0.004 0.000					
Dairy products per visitor (lb)	0.044 0.000					
Meat per visitor (lb)	-0.007 0.000					
Snacks per visitor (lb)	0.031 0.001					
Non-dairy beverages per visitor (lb)	0.011 0.000					
Non-food necessities per visitor (lb)	-0.023 0.000					
Unknown no. of visits ^b	-1.351 0.003					
<i>Panel D. Pantry type</i>						
Church	-0.224 0.005	0.413 [0.006]	0.421 [0.006]	-0.076 [0.008]	-0.621 [0.013]	
Mobile pantry or produce market	-1.351 0.003					
Nonprofit	0.246 0.003					

Notes: This table presents estimates of the interaction terms in a conditional logit model of food pantry choice (see Equations (4) and (5)). USDA store types are aggregated following Shannon (2022). Heteroskedasticity-robust standard errors appear in brackets.

7.1 Consumer Surplus and Willingness to Travel

The parameters identified in the fully interacted model capture how customer preferences vary across different demographics. Using these preferences, I compare differences in per-visit consumer surplus and willingness to travel between households living in different income quintiles, allowing me to determine how the geographic distribution of food pantries affects the value customers derive from each visit. The surplus for a household in income quintile A is given by:

$$\begin{aligned} E[CS \mid i \in A] &= E \left[\frac{1}{\alpha_i} \max_{j \in \mathcal{J}_i} \{ \mathbf{x}_{jt} \beta_i - \alpha_i D_{ij} \} \right] \\ &= E \left[\frac{1}{\alpha_i} \log \left(\sum_{j \in \mathcal{J}_i} \exp(\mathbf{x}_{jt} \beta_i - \alpha_i D_{ij}) \right) \right], \end{aligned} \quad (6)$$

where A indicates the set of households in quintile A . Equation (6) allows for variation across individuals within income quintiles, such as households within a particular income quintile with heterogeneous distance preferences depending on household size. Typical consumer surplus calculations give the difference between the highest price a customer would be willing to pay and the price they actually pay for a good or service. Because food pantries are free, this calculation gives the difference between the farthest distance a customer would be willing to travel to visit pantry j and what they actually travel to visit pantry j . In other words, consumer surplus is the number of additional miles an average household in each income quintile would be willing to travel before they were indifferent between visiting a particular food pantry, the inside option, or purchasing all meals from a grocery retailer, the outside option. Customers who derive a greater benefit from visiting a food pantry are willing to travel farther because the benefits of visiting outweigh the cost of travel. However, if a customer is more sensitive to travel distance, they may not be willing to travel as far as a customer who is less sensitive to travel distance.

Table 9 compares the consumer surplus derived by households in different income quintiles, along with the predicted number of visits and distance traveled.²⁸ The table indicates that households in all quintiles gain approximately the same amount of consumer surplus per visit; households would be willing to travel an additional 1.1 to 1.2 miles to access their chosen food pantry before they would choose the outside option of purchasing all groceries from a local retailer.

However, households in the highest-income neighborhoods are predicted to travel more than double the distance of households in the lowest-income neighborhoods. While customers in the lowest-income neighborhoods travel 3.3 miles each way and would tolerate a 1.1-mile increase (a total of 4.4 miles), those in the highest-income neighborhoods travel 7.9 miles and would still travel an additional 1.1 miles before choosing the outside option. In total, households in the highest-income

²⁸Appendix Table 5 reports the average observed visits and distance traveled. I report the predicted results to ensure the results are comparable to the counterfactual analysis in Section 7.2.

Table 9: Customer Welfare

Income quintile of census tract	Surplus from food pantries (miles)	Predicted welfare-relevant outcomes		
		Distance per regular- pantry visit (miles)	Regular-pantry visits per year ^a	Mobile-pantry visits per year ^a
1	1.12 (0.69)	3.26 (3.02)	9.40 (6.52)	1.19 (2.34)
2	1.10 (0.73)	5.86 (4.13)	9.38 (7.03)	1.52 (2.64)
3	1.15 (0.75)	6.69 (4.07)	10.08 (6.98)	1.41 (2.26)
4	1.18 (0.76)	6.99 (3.42)	11.87 (8.00)	1.21 (2.14)
5	1.14 (0.57)	7.87 (3.02)	12.27 (6.64)	1.03 (1.96)

Notes: This table compares customer surplus, travel distance, and frequency of pantry visits according to the income quintiles of the census tracts where customers live. The annual frequency of regular (mobile) pantry visits is computed as the sum of observed regular (mobile) pantry visits, divided by the number of years elapsed between the earliest and latest observed visit to a pantry of any type. Standard errors are reported in parentheses.

^a Conditional on thirty or more days elapsing between earliest and latest trip dates

neighborhoods are willing to travel 9.0 miles each way to visit their chosen food pantry. Moving from the lowest-income to the highest-income neighborhood increases willingness to travel by over 100%.

Although customers living in the highest income neighborhoods are predicted to travel farther, these households are also more sensitive to traveling longer distances than households in low-income areas. If households in high-income neighborhoods and low-income neighborhoods were visiting identical pantries the same distance from their residence, households in high-income neighborhoods would gain less surplus from each visit because of a stronger preference for nearby pantries. Therefore, for consumer surplus to remain equal across income quintiles, the utility provided by pantries within the choice set of customers in high-income areas must be substantially higher to fully offset the longer travel distance and greater travel sensitivity. The results suggest that differences in willingness to travel more likely stem from the variation in the service design of the pantries in different choice sets, rather than variation in preferences. By traveling farther, households in high-income neighborhoods can access pantries that provide enough additional benefit to offset their travel sensitivity, while households in low-income neighborhoods do not have access to pantries that provide enough benefit for them to increase their travel distance by much.

Section 5 established that there are fewer and more restrictive pantries in low-income areas than high-income areas, which led to the question of how this impacts the value that customers derive from their visit to the pantry. The demand estimation shows that customers in lower-income areas receive less value from visiting the food pantry, suggesting that the restrictions imposed on customers

in low-income areas lead to reductions in utility.

7.2 Counterfactual Simulation

In this section, I explore whether increasing funding for food pantries in low-income areas would improve aggregate welfare for food pantry customers. I reallocate funds from pantries in high-income areas to pantries in low-income areas, allowing pantries in low-income areas to open more frequently and during times that customers prefer. This exercise aims to demonstrate the existence of welfare-improving reallocations rather than to derive an optimal distribution. Evidence of such improvements suggests that the decentralized, donor-led funding equilibrium may fail to adequately serve high-need areas. These estimates serve as a benchmark for potential welfare gains under a more centralized or needs-based allocation strategy.

For this exercise, I focus on redistribution among pantries located in rural areas, meaning that pantries located in Franklin County are excluded.²⁹ Estimates in Table 7 show that customers prefer food pantries that are open on Fridays and in the evenings, motivating the redistribution of funds to allow pantries in low-income areas to increase the number of weeks they are open on a Friday or in the evenings. I focus on a subset of pantries in low-income areas that: (i) are infrequently open on Fridays or evenings and (ii) serve a higher-than-average volume of customers. These pantries would receive additional resources to open on Fridays or evenings during weeks where the pantry is already open at least one day. The funding comes from savings generated by closing food pantries in high-income areas on all Wednesdays or afternoons that they are currently open. To minimize the harm from closing the pantries, I choose a subset of pantries that serve a lower-than-average volume of customers. Opening the low-income pantries during the more popular times between 2018-2023 would increase expenditure by approximately \$595,088, and savings from the closures would allow \$614,089 to be redistributed to the lower income pantries. For details on how I estimate the predicted change in food pantry funding, see Appendix C.

Table 10 shows that pantries mostly serving customers in quintile two are open ten Fridays per year on average, while pantries mostly serving customers in quintile three are open 33 Fridays per year. Similarly, pantries serving customers in quintile two are rarely open in the evenings, at just under seven evenings per year. Pantries serving customers in quintile three and five are open between 11 and 13 evenings per year, or around one evening per month. This suggests that increasing the number of weeks pantries are open during these preferred times in low-income areas could increase welfare overall.

²⁹Recall that although all of the customers in this analysis live in the 19 rural counties that MOFC serves, some choose to visit pantries within Franklin County if they live nearby. Because I do not include Franklin County customers in my overall welfare analysis, I cannot account for how changing pantry characteristics for a pantry in Franklin County would affect the welfare of a customer in Franklin County.

Table 10: Pantry Characteristics by Income Quintile

Income Quintile	Total Pantries	Average weeks open per year	Average weeks open on a Friday per year	Average weeks open at least one evening per year
1	2	26.4 (15.6)	18.6 (12.7)	0.4 (0.74)
2	66	30.8 (20.7)	10.1 (15.6)	6.6 (12.1)
3	48	32.6 (19.5)	32.6 (13.2)	11.5 (17.4)
4	8	29.4 (22.7)	6.6 (12.0)	8.4 (13.9)
5	8	37.1 (17.0)	20.1 (21.6)	13.0 (18.5)

Notes: Table 10 breaks down the income quintile assigned to each pantry. A pantry is assigned an income quintile based on the population-weighted average income quintile of the census tracts with a population centroid within twenty miles of the food pantry. The table shows the total number of pantries assigned to each income quintile, the average number of weeks per year the pantries are open, and the average number of weeks per year the pantries are open on a Friday or in the evening. (Source: author's calculations using MOFC and ACS data.)

Table 11: Changes in Customer Welfare under Alternative Funding Distribution

Income quintile of census tract	Surplus from food pantries (miles)	Predicted welfare-relevant outcomes		
		Distance per regular pantry visit (miles)	Regular pantry visits per year ^a	Mobile pantry visits per year ^a
1	0.06 (0.12)	−0.01 (0.17)	0.42 (0.81)	−0.01 (0.02)
2	0.05 (0.15)	−0.01 (0.22)	0.45 (1.02)	−0.01 (0.03)
3	0.05 (0.14)	0.06 (0.21)	0.40 (0.97)	−0.01 (0.02)
4	−0.03 (0.08)	0.04 (0.16)	−0.28 (0.61)	0.01 (0.01)
5	−0.01 (0.03)	0.02 (0.08)	−0.15 (0.27)	0.00 (0.01)

Notes: This table reports changes in various measures of customer welfare under the counterfactual funding transfers described in Section 7.2. The annual frequency of regular (mobile) pantry visits is computed as the sum of observed regular (mobile) pantry visits, divided by the number of years elapsed between the earliest and latest observed visit to a pantry of any type. Standard errors are reported in parentheses.

^a Conditional on thirty or more days elapsing between earliest and latest trip dates

Table 11 gives the change in consumer welfare under the new regime. Consumer surplus increases by 4–5% for customers living in neighborhoods in the first through third income quintiles. Customers in quintile four experience a 2.5% decrease in consumer surplus as a result of the redistribution of resources and customers in quintile five experience less than a 1% decrease in surplus. These welfare changes are driven entirely along the intensive margin; rather than drawing new households away from the outside option, the reallocation allows existing pantry users to increase their predicted visit frequency by providing an improved selection of distribution times. Improving welfare for the customers in low-income areas is relatively low cost at around \$600,000 over a six-year period and only involves small adjustments in pantry characteristics. Larger adjustments, such as increasing distribution hours or reducing visit restrictions could result in even greater welfare improvements. Ultimately, identifying any welfare-improving funding structure suggests that relying on donors to determine which food pantries receive the most resources does not provide the highest level of welfare for food pantry customers. Instead, these results show that small changes in the funding distribution, guided by customer rather than donor preferences, can lead to welfare gains for households in the lowest-income areas, highlighting the potential for more equitable resource allocation.

8 Conclusion

This paper analyzes the relationship between economic resources and the availability of charitable services in a geographic area. I begin by documenting differences in the availability of food pantries between households in high- and low-income neighborhoods. Households in high-income neighborhoods live near over 40 more food pantries than households in low-income neighborhoods, even after adjusting for population and access to urban areas. Customers in high-income areas also live near more pantries with a desirable service design, such as flexible visit times.

The descriptive analysis points toward differences in food pantry access that could lead to disparities in consumer welfare. I then estimate the consumer surplus derived by households in different income quintiles. The results show that households in high-income neighborhoods are willing to travel farther to reach a food pantry than households in low-income neighborhoods even though households in higher-income areas experience more disutility from travel. This implies that customers in high-income areas are willing to pay a higher price for a higher quality visit. Because five times as many customers live in low-income neighborhoods compared to high-income neighborhoods, improving access for customers in low-income neighborhoods could lead to significant welfare improvements for a large portion of the population.

The analysis of the status quo points toward variation in food pantry quality that favors customers in high-income neighborhoods. Potential donors typically earn more than \$75,000 per year, according to the MOFC food pantries, and research shows that charitable giving and volunteerism increase with

education (Jones 2023). Therefore, the demographics of the high-income areas are consistent with the profile of donors and volunteers to food pantries, suggesting that local-giving may be driving the disparities in food pantry access. In a counterfactual analysis, I show that welfare-improving funding structures exist with small adjustments to the amount of funding each pantry receives. I simulate opening high-volume pantries serving low-income neighborhoods on popular days and closing low-volume pantries in high-income neighborhoods on less popular days, according to preferences derived using the random utility model. I find that the new funding structure increases consumer surplus for customers in quintiles one, two, and three between 4–5%. Byrne and Just (2023) show that each pantry visit is worth between \$40 and \$60, so a 4-5% increase adds \$2-3 per visit, or approximately the average cost of an additional meal (Feeding America 2025). MOFC estimates that households in their region miss between 3 and 4 meals per week, and so small welfare gains can begin to close the meal gap.

More broadly, this study suggests that relying on decentralized funding to provide charitable social services inherently generates geographic inequality. Households in wealthier areas receive superior care, while those in poorer areas face significant support gaps. Policies that assume charitable organizations can seamlessly step in to replace federal safety net cuts fail to account for the spatial mismatch between where resources are raised and where they are needed. This paper demonstrates that the charities with the most robust infrastructure are systematically concentrated in the neighborhoods that need them least.

While food pantries in higher-income neighborhoods provide more value per visit, access to food pantries improves welfare in all areas, regardless of neighborhood income. Private food assistance provides a small, but meaningful, amount of support to households in need of food, and in speaking to food pantries across the MOFC service area, all pantries felt they were falling short of meeting the demand in their area. Redistributing resources from high-income areas to low-income areas would improve welfare overall for pantry customers, but would ultimately decrease welfare for some needy households living in high-income areas. In addition, donors likely gain more utility from supporting their neighbors, and therefore redistributing funding would harm donors or could lead to decreases in overall giving.

Alternative solutions to disparities in high-quality food pantry access, and charitable service access in general, could involve increasing outside funding, such as government funding or grant funding, for pantries in low-income areas. Without paid staff, small, all-volunteer organizations may not have the time or expertise required to apply for large grants that could significantly improve welfare in their region. Improvements in data collection, such as the data collected by MOFC, could allow policymakers to identify organizations poised for high-impact improvements without requiring grant-writing skills from those organizations.

Another approach is to allow intermediary organizations, like food banks, to identify underfunded

organizations and redistribute the resources accordingly. For example, MOFC currently identifies organizations in underserved areas that are capable of expansion and supports them with marketing materials and additional produce to increase charitable food access in their service area. The Greater Chicago Food Depository, a large food bank in the Chicago area, also implemented a creative solution to expensive shipping costs; the food bank provided small grants to partner food pantries that were capable of storing large volumes of food. The grants allowed the pantries to purchase refrigerated vehicles that they could then use to travel to the food bank to pick up food orders for other pantries in their region. The smaller pantries, then, only needed to travel to the grant-funded pantry to pick up food, rather than pay to have the food shipped all the way from the bank.

In addition to showing that relying on donors to fund charitable social services leads to lower levels of aggregate welfare than alternative funding structures, I also contribute to the small literature on food pantries by describing the preferences of households relying on food pantries for some of their meals. Households prefer nonprofit over church-based pantries, favor distribution times on Fridays and in the mornings or evenings, and prefer pantries offering more dairy and snacks per visit. Preferences for food pantry service design vary by the customer's neighborhood income and also by the customer's household size. Finally, the local food environment also influences the benefit derived from visiting a food pantry; customers with a higher concentration of supermarkets or local retailers near their home derive more benefit from the outside option of shopping at a grocery retailer, suggesting that vibrant grocery shopping options may draw food pantry customers toward the outside option and away from visiting the pantry. On the other hand, access to more small retailers, like convenience stores, does not shift the value of the outside option for food pantry customers.

To my knowledge, this paper is the first quantitative analysis of demand for food pantries using multi-county data on food pantry utilization. However, this paper is limited to households who select into visiting a food pantry. By focusing on the customers who choose to visit, this analysis cannot account for the preferences of those who do not visit the pantry and therefore cannot inform policy decisions to draw more individuals into the food pantry system. In addition, this paper assumes that the cost of visiting a food pantry is the Euclidean distance traveled to the food pantry. Although the Euclidean distance is likely correlated with travel times, the actual distance traveled to each food pantry is unknown and almost certainly farther than the Euclidean distance. Therefore my estimates are a lower bound for the travel costs associated with visiting a pantry. Further, I am unable to account for the variation in wait times that customers face when visiting a pantry, which could increase the cost of visiting. For these reasons, I generate a lower bound to the value of visiting a food pantry in miles, rather than dollars. Byrne and Just (2023) find that food pantries provide an annual benefit of \$600 to \$1,000 per customer; my work complements that research by showing that the benefits provided by pantries likely vary by geographic location.

In 2016, MOFC estimated that charitable food provided 5% of the meals needed by low-income

households in the 20-county region, while government provided meals, including SNAP, WIC, TANF, and free/reduced-price lunch, provided 29%. Feeding America reports that the personal budget shortfall that food insecure households face increased by 55% between 2020 and 2023 (Feeding America 2025). However, Feeding America (2025) also shows that two out of five households facing food insecurity do not qualify for SNAP benefits, and therefore these households may depend more on charitable services. Recent cuts to federal programs will make it more difficult for households to receive food benefits along with other federal social services. State-level cuts to food banks and food pantries, such as recent cuts in Ohio (Sollinger 2025), will restrict the ability of charitable services to respond to the increase in need. This research indicates that while charitable services provide an important resource for low-income households, variation in local funding and support leads to capacity constraints for some organizations. Alternative funding structures that use existing resources and prioritize customer preferences would improve customer welfare overall.

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Supplementary Appendix

A Tables

Appendix Table 1: Food Pantry Orders: Branches without MOFC

	<i>Panel A. Total quantity ordered</i>
Weight (lb)	151,664.4 (226,771.5)
Price (\$)	13,464.4 (18,256.5)
	<i>Panel B. Quantity ordered by food category (%)</i>
Bread	3.5 (5.9)
Shelf-stable	29.7 (16.7)
Fresh foods	44.8 (24.1)
Dairy	6.0 (4.9)
Meat	12.3 (8.5)
Snacks	10.1 (6.9)
Non-dairy beverages	7.6 (7.6)
Non-food necessities	4.3 (4.8)
Observations (pantry-year)	24

Notes: This table shows the amount of food in pounds that all branch organizations ordered, excluding the two locations operated by the Mid-Ohio Food Collective. (Source: author's calculations using MOFC data.)

Appendix Table 2: Food Pantry Orders by Nonprofit Type in 2022

	<i>Panel A. Total quantity ordered</i>			
	990	990-EZ	990-N	Nonfiler
Weight (lb)	397,077.3 (501,252.3)	187,355.6 (360,267.5)	187,443.9 (462,234.7)	149,856.8 (264,332.3)
Price (\$)	15,371.1 (20,110.1)	10,030.8 (18,379.42)	5,036.4 (7,286.1)	5,351.2 (6,555.2)
	<i>Panel B. Quantity ordered by food category (lb)</i>			
	990	990-EZ	990-N	Nonfiler
Bread	6.1 (6.7)	7.9 (11.8)	4.7 (7.8)	3.4 (6.2)
Shelf-stable	17.8 (11.9)	27.0 (18.2)	28.3 (18.4)	26.4 (13.3)
Fresh foods	57.7 (21.3)	46.7 (22.0)	38.9 (18.8)	39.6 (20.6)
Dairy	6.5 (3.9)	6.3 (4.4)	5.9 (5.0)	5.7 (3.9)
Meat	12.2 (9.0)	9.8 (6.0)	14.3 (6.1)	14.5 (9.5)
Snacks	8.3 (6.2)	10.0 (6.4)	11.8 (6.5)	12.8 (7.2)
Non-dairy beverages	5.0 (4.7)	6.0 (5.6)	6.1 (5.6)	9.5 (8.1)
Non-food necessities	4.9 (8.0)	4.8 (4.6)	5.7 (8.0)	5.7 (6.2)
Observations (pantry-year)	158	45	75	101

Notes: This table breaks down food pantry orders from nonprofit organizations by nonprofit tax-filing type. I exclude 2023 as not all organizations' tax filings were available at the time of writing this paper. (Source: author's calculations using MOFC data.)

Appendix Table 3: Household Size and Retail Food Availability by Tract Income Quintile

	Income quintile of Customer's census tract				
	1	2	3	4	5
<i>Panel A. Household size</i>					
Customer-reported household size	2.5 (1.9)	2.6 (1.8)	2.6 (1.8)	2.7 (1.9)	3.0 (2.0)
<i>Panel B. Retail food availability</i>					
Number nearby grocers	1.3 (0.8)	1.8 (4.1)	0.7 (0.8)	0.6 (0.8)	2.1 (4.9)
Number nearby local food stores	0.9 (0.9)	0.7 (0.9)	0.4 (0.7)	0.4 (0.7)	0.3 (0.6)
Number nearby small retailers	19.4 (13.8)	18.0 (15.6)	11.8 (10.5)	8.0 (9.1)	12.3 (16.6)
Number nearby supermarkets	3.9 (3.3)	3.5 (3.2)	2.1 (2.4)	1.5 (2.2)	2.7 (4.1)

Notes: This table presents summary statistics about household size (as reported by MOFC customers), as well as the availability of retail food options within five miles of a customer's home, according to the income quintile of customers' census tracts. (Sources: MOFC and USDA).

B Figures

Appendix Table 4: Estimated Base Coefficients (β_k 's and α)

	Point estimate	Standard error
<i>Panel A. Travel cost</i>		
Distance $\times -1$	0.223	0.001
<i>Panel B. Pantry flexibility</i>		
No cap on groceries obtained	-0.664	0.006
Limit one visit per week	-1.800	0.006
Open mornings	0.294	0.003
Open afternoons	-0.121	0.005
Open evenings	0.288	0.003
Unknown open hours ^b	-0.707	0.007
Open Sundays	0.068	0.006
Open Mondays	-0.328	0.002
Open Tuesdays	-0.072	0.003
Open Wednesdays	0.167	0.003
Open Thursdays	-0.062	0.003
Open Fridays	0.320	0.004
Open Saturdays	-0.090	0.004
<i>Panel C. Food per customer</i>		
Bread per visitor (lb)	0.011	0.000
Shelf-stable food per visitor (lb)	-0.025	0.000
Dairy products per visitor (lb)	0.044	0.000
Fresh produce per visitor (lb)	-0.004	0.000
Snack food per visitor (lb)	0.031	0.001
Meat per visitor (lb)	-0.007	0.000
Non-dairy beverages per visitor (lb)	0.011	0.000
Non-food necessities per visitor (lb)	-0.023	0.000
Unknown no. of visits ^a	-1.351	0.003
<i>Panel D. Pantry type</i>		
Church	-0.224	0.005
Mobile pantry or produce market	-1.351	0.003
Nonprofit	0.246	0.003
<i>Panel E. Retail availability</i>		
Outside option $\times$ no. nearby grocers	-0.015	0.000
Outside option $\times$ no. nearby local food stores	0.104	0.002
Outside option $\times$ no. nearby small retailers	-0.004	0.000
Outside option $\times$ no. nearby supermarkets	0.104	0.001

Notes: This table presents base coefficient estimates for a conditional logit model of food pantry choice (see Equation (3)). Concerning pantry type (Panel D), the omitted type of pantry is branch locations. The rightmost column reports heteroskedasticity-robust standard errors.

^a When the number of weekly visits is unknown, this variable is defined as one and the per-customer food variables are set equal to zero.

^b Open hours cannot be inferred because pantry does not report visit times.

Appendix Table 5: Observed Travel Distance and Pantry Visit Frequency

Income quintile of census tract	Distance per regular pantry visit (miles)	Regular pantry visits per year ^a	Mobile pantry visits per year ^a
1	2.8 (4.0)	9.0 (9.1)	1.3 (4.0)
2	4.3 (4.4)	9.3 (9.6)	1.5 (4.3)
3	5.8 (5.2)	10.0 (9.7)	1.1 (3.7)
4	6.1 (4.9)	11.4 (10.8)	1.1 (4.2)
5	7.5 (5.0)	12.5 (11.7)	1.2 (4.9)

Notes: This table compares observed travel distance, and frequency of pantry visits according to the income quintiles of the census tracts where customers live. The annual frequency of regular (mobile) pantry visits is computed as the sum of observed regular (mobile) pantry visits, divided by the number of years elapsed between the earliest and latest observed visit to a pantry of any type. Standard errors are reported in parentheses.

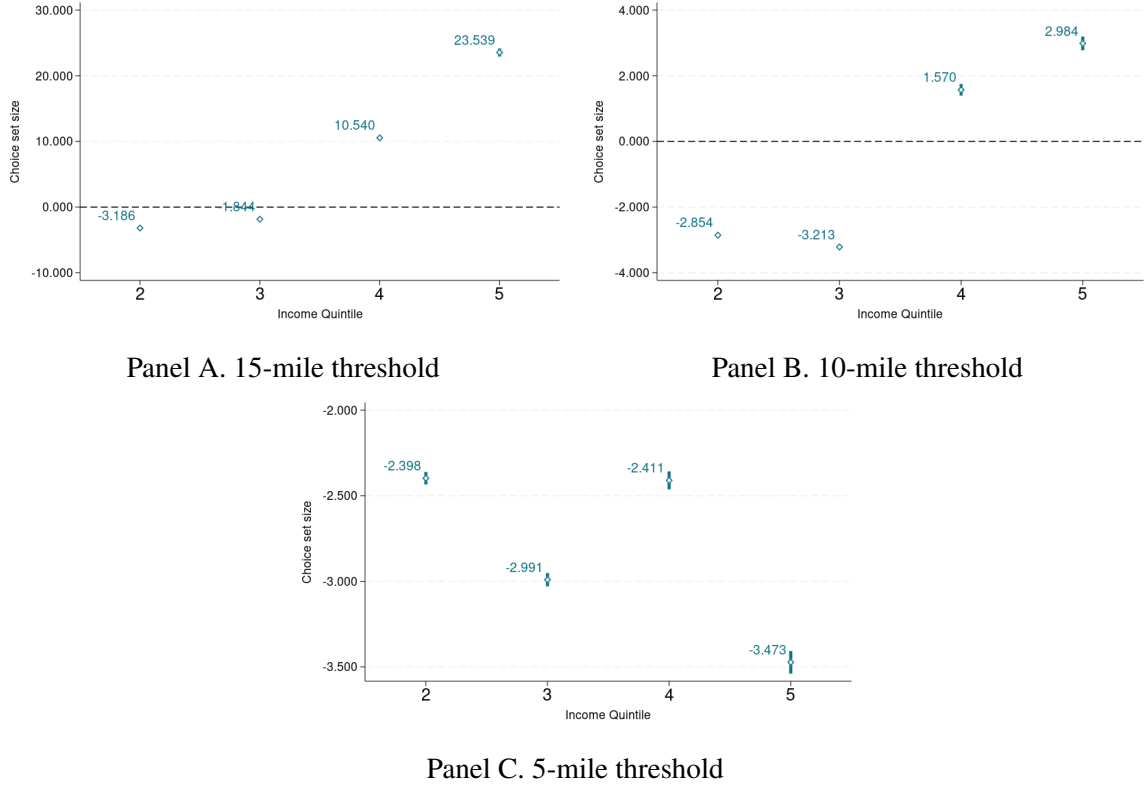
^a Conditional on thirty or more days elapsing between earliest and latest trip dates.

C Predicted Funding Changes

To estimate the cost of opening food pantries in low-income areas and the savings from closing food pantries in high-income areas, I first identify the *pantry* income quintile as the income quintile of the people likely served by the pantry and/or the income quintile of potential supporters of the pantry. Specifically, I use the weighted average of the income quintiles with a population centroid within 20 miles of the pantry. By this definition, there are two pantries that primarily serve customers in quintile one, 66 pantries primarily serving customers in quintile two, 48 primarily serving quintile three, and eight primarily serving both quintiles four and five. Excluding Franklin county removes 135 pantries that rural customers chose, all located in quintiles three or above.

Food pantries face both fixed and variable costs when providing food to the community. I assume that the amount of money each pantry spends on food in a given month represents the variable costs that pantries face. I treat other costs, such as staff and rent, as fixed and I assume these costs do not change if the pantry opens or closes on additional days. I exclude fixed costs in the cost-per-visit.

To calculate the cost-per-visit, I divide the total amount spent on food by pantry j in month t by the total number of visits to pantry j that occurred in month t . I multiply the number of weekly visits by the cost-per-visit to estimate the weekly cost of serving customers. For pantries open on Fridays or in the evening, I calculate the share of their visits that take place on Fridays/evenings. I then predict the share of visits that would occur at pantries not currently open on Fridays or evenings using a beta



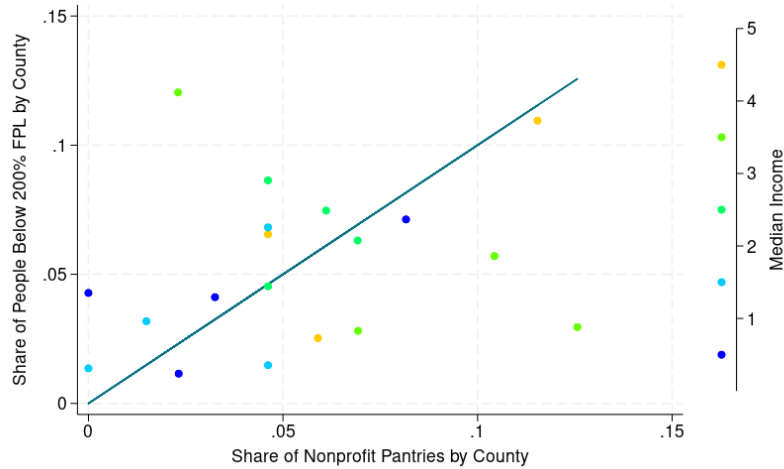
Appendix Figure 1: Variation in Total Options in Choice Set by Income Group

Notes: This figure plots the β_1 coefficients and 95 percent confidence intervals from Equation 2 using alternative definitions of the food pantry choice set. Each graph plots the change in the count of food pantries in the choice set of a customer living in each income quintile, relative to the count of pantries in the choice set of a household living in lowest income quintile neighborhood.

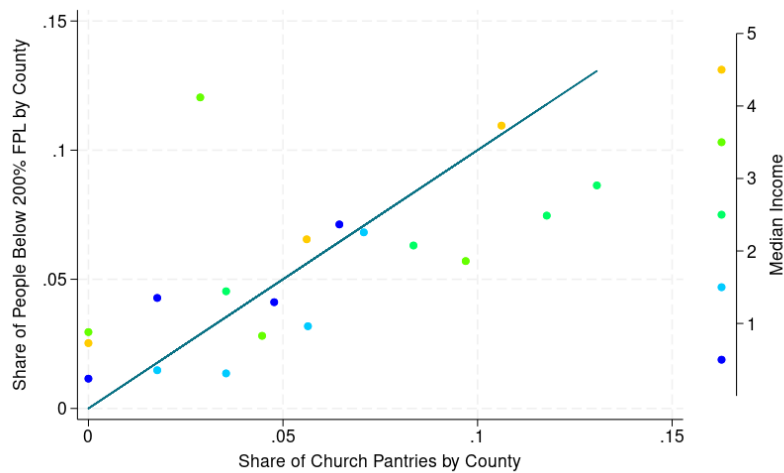
regression, represented by:

$$g(y_{jt}) = \mathbf{x}_{jt}\boldsymbol{\theta} + \tau_t + \gamma_j, \quad (7)$$

where $y_j \in \{0, 1\}$ is the share of visits on a Friday or in the evenings. This specification does not include pantries that are not open on Fridays or in the evenings or pantries that are only open on Fridays or in the evening in a given week. I also include a vector of pantry characteristics, $\mathbf{x}_j$, including indicators for pantry type (church, or nonprofit), day-of-the week the pantry is open, and time-of-day the pantry is open. I include the pantry-census-tract population of individuals between ages 36 and 64, which MOFC reports as the age range where most volunteers fall, to control for the availability of volunteer labor for a particular pantry. I include year and month fixed effects, τ_t , to account for unobservable patterns in pantry utilization related to the year and month the pantry is open. Finally, I include pantry fixed effects to account for unobservable, time-invariant pantry characteristics.



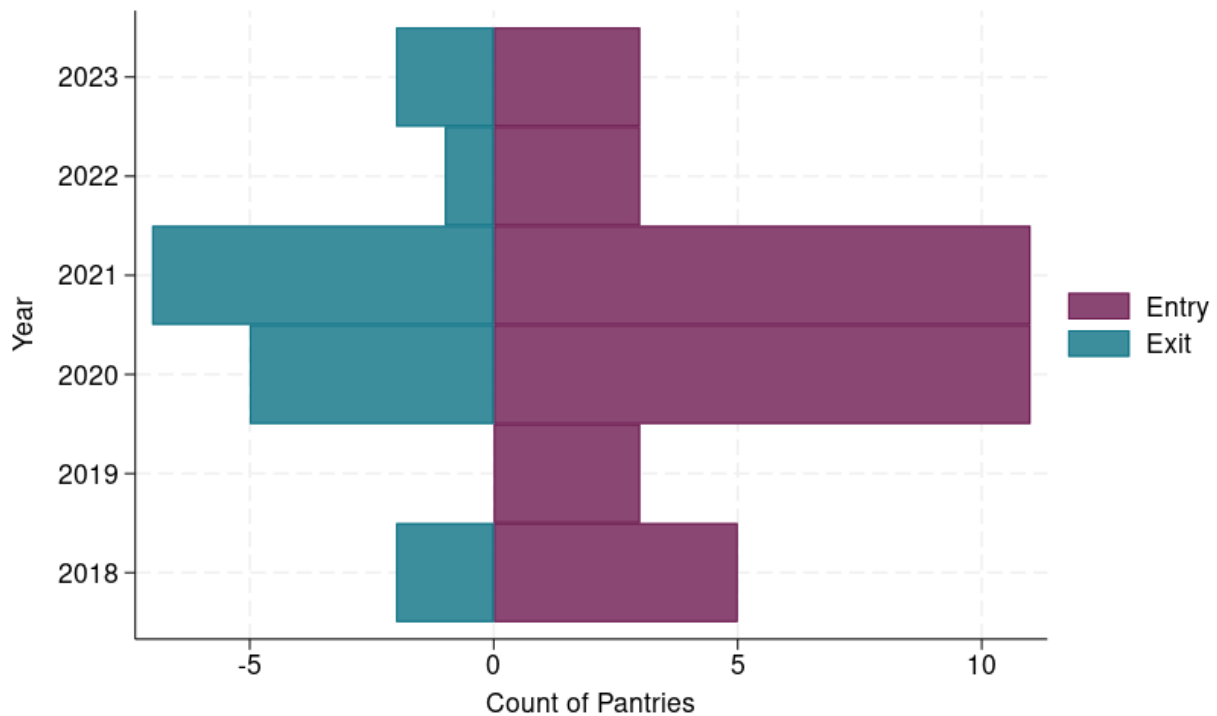
Panel A. Share of Nonprofits and Share of Pantry-Eligible Households



Panel B. Share of Churches and Share of Pantry-Eligible Households

Appendix Figure 2: Distribution of Church and Nonprofit Pantries by County

Notes: The figure plots the share of households in the MOFC region below the 200% of the federal poverty line residing in each county against the share of nonprofit or church-based food pantries in each county, and a 45 degree line. Points above the line have a disproportionately high share of individuals eligible for food pantry services. Points below the line have a disproportionately high share of food pantries. The points are colored according to the income quintile of the county.



Appendix Figure 3: Entry and Exit of Pantries by Year

Notes: This figure shows the total number of pantries that enter or exit the sample by year.

I calculate the estimated cost increase by multiplying the predicted weekly visits on Fridays or evenings by the pantry's weekly cost. Similarly, I calculate the actual cost-per-week for pantries that are open on Wednesdays or in the afternoon by multiplying the actual share of weekly visits that occur during these times by the pantry's weekly costs. I then choose the subset of pantries in low-income areas that are (i) infrequently open on Fridays (evenings), defined as open on a Friday (evening) for less than half of the weeks that the pantry is open; and (ii) a pantry with a higher-than-average volume of customers, relative to the other low-income pantries that are infrequently open on Fridays or in the evenings. This includes 16 food pantries eligible for Friday openings, half of which are church pantries, seven small nonprofit organizations, and one branch location. 18 pantries are eligible for evening openings, including eight churches, three branches, six small nonprofits, and one mid-sized nonprofit. Because the pantries are largely churches and small nonprofits, which are typically all-volunteer operations, the labor costs associated with the additional hours would involve recruiting more volunteers rather than hiring new staff. Though the list of pantries that would open on Fridays and pantries that would open during the evenings are distinct, eight pantries are on both lists.

I calculate the expected savings that could be redistributed to low-income pantries if pantries in

quintiles three, four, and five that are frequently open on Fridays or in the evenings were to close on Wednesdays and during the afternoons. I choose to focus on pantries that are frequently open on Fridays or evenings because this would allow customers to continue to visit the pantry on the most popular day and time. For pantries that would close, I focus on low-volume pantries, or pantries that serve a lower-than-average volume of customers. Eight pantries would experience afternoon closures in the counterfactual scenario and eleven would experience Wednesday closures. Only one pantry appears on both lists.